OpenText™ Service Manager

Create stored queries from the Query Maintenance form

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Create stored queries from the Query Maintenance form

You can add, edit or run stored queries from the Query window, which is accessible from the form in Database Manager or from the Application search form. Data entered into the form is automatically entered into the Query window to form the query. Depending on your permissions, you can edit and run that query, or select a query from a list of stored queries.

1. Click **Tailoring** > **Tailoring Tools** > **Stored Queries**.
   A blank Stored Query Maintenance form opens.
2. Click **Search**.
   A list of existing queries opens.
3. You can copy an existing query, create a brand new query, or create a new query from an existing query.

   Note

   Note In Service Manager, you can use field name information in field help to identify the field names you want to use in your stored queries. To display field name information in field help, make sure one of the following conditions is fulfilled:

   - In the Windows client, the "Show context-sensitive help debug information" preference is enabled.
   - In the Web client, "viewcontexthelp=true" is appended to the URL before you log in.

4. Click **Save** to save any changes.
   Caution: If you are creating a new stored query from an existing query, make sure you click Add instead of save. If you click save, you will replace the existing stored query with the new stored query you are attempting to add.
5. Click **Add** to add a new query.
   You receive a message that states the record has been added to the querystored file.

Related topics

Stored queries
Using stored queries in display objects
Using stored queries in scripts
Menu option searches

1. Form creation

These topics focus on creating and customizing forms within the system. The tools that are used in form creation are:

- Forms Designer — alters the data in one or more dependent files to match changes made to data in a source file.
- Dynamic View Dependencies (DVD) — applies values, sets mandatory fields, and validates at the table level.
- Format Control — defines a field's valid values.

Related topics

Tailoring
Forms Designer
Creating and editing forms
Forms Designer best practices
Building accessible forms
Dynamic View Dependencies
Format Control
Access Forms Designer
Create a form using the Form Wizard

Add a dynamic form to a form
Forms Designer controls and tools

1.1. Forms Designer

Forms Designer is the utility you use to design, create, and update Service Manager forms. This utility gives access to a drawing canvas upon which you construct your forms, a design toolbar from which you access controls and layout objects, and a properties view that you use to set attributes for each control or layout object. Forms Designer provides a Form Wizard that you can use to automatically create forms based on a particular database dictionary.

Related topics

Form creation
Creating and editing forms
Forms Designer best practices
Building accessible forms
Dynamic View Dependencies
Format Control
Access Forms Designer
Create a form using the Form Wizard
Forms Designer controls and tools

1.1.1. Access Forms Designer

Applies to User roles: System Administrator

You can open Forms Designer using one of the following methods:

- To access Forms Designer from the System Navigator, click **Tailoring** > **Forms Designer**.
- To access Forms Designer from the Service Manager command line, type `fd` and press Enter.

From this screen, you can create a new form or search for an existing form.

Related topics

Form creation
Forms Designer
Creating and editing forms
Using the drawing canvas
Forms Designer properties view
Setting properties
Forms Designer best practices
Building accessible forms
Dynamic View Dependencies
Format Control
Create a form using the Form Wizard
Update a form
Add a dynamic form to a form
Forms Designer controls and tools

1.1.1.1. Creating and editing forms

The Form Wizard enables you to create forms based on a particular Service Manager table. You can automatically create both table–type record lists and single–record display forms. You can only use the Form Wizard to create new forms; it can't modify existing forms.

Forms Designer design mode contains a **Drawing Canvas**, **Properties view**, and **Design toolbar**. You use them to create, design and modify Service Manager forms.

Forms Designer enables you to use the forms you create in three basic ways. A form can be used as a stand-alone, as a subform embedded in a parent form, or as a pop-up.

Related topics

Form creation
Forms Designer
Using pop-ups
Using the drawing canvas
Forms Designer best practices
Access Forms Designer
Create a form using the Form Wizard
Add a dynamic form to a form
Add a subform to a form
Forms Designer controls and tools

1.1.1.1.1. Create a form using the Form Wizard

Applies to User roles: System Administrator

You use the Form Wizard to create a new form for an existing database table. You must use the Windows client to create a new form in Forms Designer.

To create a form using the Form Wizard:

1. Click **Tailoring** > **Forms Designer** in the System Navigator, or type `fd` on the Service Manager command line and then press **Enter**.
2. Click **New**.
3. Click **Yes** to use the Form Wizard.
4. Type the name of the new form and click **OK**.
5. Type or select the name of the table you want to associate with the form, and click **OK**.
6. You can create a form that displays a single record or a form that displays a list of records. Select the type of form you want to create and click **OK**.
7. Select a size for the form.
8. Change the `true` to `false` in the Show (t/f) column if you don't want a particular field included on the completed form.
9. Change the text in the Field Label column to update the label on the form.
10. Click **Proceed**. The new form opens in Design mode.
11. Add a field to the new form, as follows:
    1. Drag the control to its approximate location, select the field, and then modify its Properties.
    2. To modify a field, select it and then modify the value of the Property you want to adjust.
    3. To delete a field, select it and press **Delete**.
    4. Click **Web Preview** to see what the form will look like in the Web client.
    5. Click **OK** to see what the form will look like in the Windows client.
    6. Click **OK** to save your changes.
12. Open the form using Database Manager to test your changes:
    1. Click **Tailoring** > **Database Manager**.
    2. Click **Search**. Service Manager opens a list of forms.
    3. Click the form you just created to open it.

Related topics

Form creation
Forms Designer
Creating and editing forms
Using the drawing canvas
Forms Designer properties view
Setting properties
Forms Designer best practices
Building accessible forms

Proprietary records
Access Forms Designer
Update a form
Add a dynamic form to a form
Forms Designer controls and tools
HTML Editor controls
HTML Editor keyboard shortcuts

1.1.1.1.2. Using the drawing canvas

When you access Form Designer to create a new form, you are provided with a drawing canvas on which you build the form.

To work effectively with the canvas, you should know the following:

- The structure of the canvas.
- How to size your forms on the canvas.
- How to position multiple objects.

The canvas contains a grid to which all design objects are referenced. Click **Grid Preview** to see the grid.

- The grid coordinates start at (0,0) in the top left corner of the canvas. The X (horizontal) coordinates increase as you move to the right. The Y (vertical) coordinates increase as you move down.
- There is approximately a 3 to 1 ratio between X units and Y units. For example, 30 X units cover approximately the same amount of space as 10 Y units.
- For best results, try to use even X Y coordinates for objects you place on the canvas. The space required to display one character is 2x2 and, overall, the client displays better when you use even coordinates.

Related topics

Form creation
Forms Designer
Creating and editing forms
Forms Designer properties view
Setting properties
Forms Designer best practices
Building accessible forms
Access Forms Designer
Create a form using the Form Wizard
Update a form
Forms Designer controls and tools

1.1.1.1.3. Forms Designer properties view

The Properties view enables you to examine and set the property value for each object on your form. The values in the property fields determine how the object appears on the form, and how the object interrelates with the user and with the database.

The Properties view displays a window containing two columns, a property column and a value column. You can click the Property tab to sort the properties alphabetically or by category.

Like any view in the client, you can undock the properties view and move it to anywhere on your desktop.

Note: You can turn on the properties view and use it outside of Forms Designer to determine form field properties for any object in focus. However, since labels can't gain focus, you can't view properties for labels outside of Forms Designer.

The Properties view includes property specific in–place editors for changing values. There are six types of property editors:

- Text
- Boolean (checkbox)
- Color swatches
- Comboboxes
- Lists (with editor window)

- Images preview

Related topics

Form creation
Forms Designer
Creating and editing forms
Using the drawing canvas
Setting properties
Building accessible forms
Access Forms Designer
Create a form using the Form Wizard
Forms Designer controls and tools

1.1.1.1.4. Setting properties

When you select a control object, the properties view displays the type of control you have selected and the properties that apply to that object. Some properties are common to all control or layout objects, while others are object specific. The Forms Designer online Help contains a description of each control and its corresponding properties.

You can use the dynamic view dependencies feature to conditionally display some controls or conditionally change how some controls appear on a form.

Properties for a design object can be divided into the following categories:

- Common properties
- Font properties
- Box properties
- Object specific properties

Related topics

Form creation
Forms Designer
Creating and editing forms
Using the drawing canvas
Forms Designer properties view
Forms Designer best practices
Building accessible forms
Access Forms Designer
Create a form using the Form Wizard
Forms Designer controls and tools

1.1.1.1.5. Enabling HTML in forms

Forms Designer has an HTML Editor and an HTML Viewer that you can add to forms.

- The **HTML Editor** enables users to add and edit HTML text in Service Manager forms. It uses standard HTML formatting.
- The **HTML Viewer** enables users to view (but not edit) the formatted HTML text entered by using the HTML Editor.

Using the HTML Editor

To add HTML, type or paste text into the HTML Editor and format it as desired. The Knowledge Management and Knowledge Management Extended toolbars enable you to switch between source and WYSIWYG mode. If you have a Knowledge Management toolbar, you can view the HTML source code and edit it directly.



Note

While in WYSIWYG mode, the editor makes choices about how to apply the tags you select. If you want more control over how the tags are applied, ask your administrator for a Knowledge Management toolbar and use source mode for editing.

To add an image, save the image locally, upload it as an attachment, and then add a `<img>` tag that points to the attachment in the HTML editor. In the web client, you can also use the copy and paste function to paste the image into the HTML Editor directly. The image is uploaded as an attachment automatically, and a `<img>` tag containing a link to the image is displayed in the HTML Editor.



Note

Out-of-box, the copy and paste function is only available in Knowledge Management. To enable this functionality for other modules, you must meet the following prerequisites:

- You are running the Service Manager 9.52 (or later) web client and server.
- The form contains an Attachment Container control.
- The Insert Image Option property is defined for the HTML Editor control on the form.

Currently, this functionality is unsupported in HTML email.

In the Windows client and web client, press Enter for a new paragraph `<p>` and press Shift+Enter for a line break `<br>`. To exit a numbered or bulleted list and start a new paragraph `<p>`, you can press Enter twice.

The HTML Viewer will render valid HTML, but you must ensure that the HTML you enter is valid and is properly integrated into the HTML document. For example:

- If you paste in HTML containing a JavaScript requiring a file provided by another application, that JavaScript will not work in the HTML Viewer.
- If you copy an HTML form with buttons, the form will render, but the buttons will not work in the HTML Viewer.

In the Windows client, if you paste text from a Microsoft Word document into the HTML Editor, the editor opens a window that allows you to select options to clean the HTML work created.

Add an HTML Editor or Viewer to a form

Applies to User Roles: System Administrator

To Add an HTML Editor or Viewer to a form:

1. Click **Tailoring** > **Forms Designer** in the System Navigator, or type `fd` on the Service Manager command line and then press **Enter**.
2. Type a form name and click **Search**.
   Service Manager opens a list of forms. Click the form you want to update to open it.
3. Click **Design** to open design mode.
4. Drag the HTML Editor icon to the desired location.
   Or
   Drag the HTML Viewer icon to the desired location.
5. Type the name in the Value text box of the Name property.
6. Specify the field or variable in which to store the HTML in the Value text box of the Input property.
7. Specify the height, width, and other properties of the HTML Editor or viewer, as desired.
8. Click **Web Preview** to see what the form will look like in the Web client.
9. Click **OK** to see what the form will look like in the Windows client.
10. Click **OK** to save your changes.
11. Open the form using Database Manager to test your changes:
    1. Click **Tailoring** > **Database Manager**
    2. Click **Search**. Service Manager opens a list of forms.
    3. Click the form you just created to open it.

Enable HTML Editor allowlist

The HTML Editor enables users to visually add and edit rich text content formatted with HTML tags. Though the HTML Editor is powerful, web sites may be abused without a proper security control. Therefore you can use the HTML Editor allowlist to define a list of allowed HTML elements.

Note

If your Service Manager platform components are upgraded to 9.81 or higher version, while your application version is lower than SM9.81, you need to manually add the `<s>`, `<a rel>`, and `<span contenteditable>` attributes to HTML Editor allowlist, so that the "Strikethrough", "Insert link", and "Insert Field" functions can work properly.

To enable the HTML Editor allowlist and define a list of allowed HTML elements, follow these steps:

1. Sign in to Service Manager as a system administrator.
2. Click **Tailoring** > **Allowlist** > **HTML Editor Allowlist**.
3. Select the **Enable HTML Editor Allowlist** checkbox. By default, this checkbox is checked.
4. Define a list of allowed HTML Tags and attributes in the **Allowed Tags/Attributes** section. For example, add script as an allowed tag so that you are able to input the `<script>` tag by using the HTML Editor.
5. Define a list of allowed URL protocols for some elements in the **Allowed URL Protocols** section. For example, add a as an allowed tag, specify href as the attribute, and then specify one or more protocols that are used by the href attribute, such as http, https, mailto, and ftp.
6. Click **Save** and **OK**.
7. Do either of the following to make the customized HTML Editor allowlist effective.
   - Restart the web application server if you are working with the Service Manager web client.
   - Sign out and then sign in to the Service Manager Windows client again.

   Note: The **Journal Updates Viewer** and **Activities Viewer** widgets in the Activities tab will get the allowlist from cached data. So after the allowlist is updated, it will take 5 minutes to refresh the cached data and reflect the updated allowlist.

After the HTML Editor allowlist became effective, you are only allowed to input the elements as listed in the allowlist when editing HTML contents using the HTML Editor. The system automatically validates your inputs and displays some warning messages if there are tags, attributes or protocols that violate the allowlist. For example:

`The HTML document contains some tags <script>,<javascript> that aren't allowed by the system. Contact your Service Manager system administrator for assistance.`

`The HTML document contains some attributes for tags (<tag attribute...>) <div onlick>,<p onerror> that aren't allowed by the system. Contact your Service Manager system administrator for assistance.`

`The HTML document contains some values in the attributes of tags (<tag attribute>) <img src>,<a href> that aren't allowed by the system. Contact your Service Manager system administrator for assistance.`

You must review your input and avoid using these invalid tags, attributes, or protocols. Alternatively, you can ask your Service Manager system administrator to include these tags, attributes or protocols to the allowlist. Otherwise, the system automatically removes these invalid elements from your input after you click **Save** in the HTML editor.

HTML Editor controls

The HTML Editor has the following formatting controls. Their availability depends on the Keyboard property selected in the HTML Editor control on the form you are using.

Icon	Name	Function
	Bold	Makes the selected text bold.
	Italic	Makes the selected text italic.
	Underline	Underlines the selected text.
	Strikethrough	Strikes through the selected text.
	Text Color	Makes the selected text the specified color.
	Background Color	Sets the background of the selected text to the specified color.
	Font Size	Sets the selected text to the specified font size.
	Clear Formatting	Removes formatting from the selected text.
	More Text	Select more text options: Subscript, Superscript, or Font Family.
	Subscript	To lower the text below the line of text.
	Superscript	To raise the selected text above the line of text.
	Font Family	Sets the selected text to the specified font.
	Paragraph Format	Sets the selected paragraphs to the specified format.
	Insert/Remove Ordered List	Makes the selected paragraphs into a ordered list.
	Insert/Remove Bulleted List	Makes the selected paragraphs into a bulleted list.
	Decrease Indent	Decreases the indent of the selected paragraph.
	Increase Indent	Increases the indent of the selected paragraph.
	Align Left	Left-aligns the selected paragraphs.
	Center	Center the selected paragraphs.
	Align Right	Right-aligns the selected paragraphs.

	More Paragraph	Select more paragraph options: Quote, Align Justify, or Line Height.
	Quote	Creates an indented quote style.
	Align Justify	Fully justifies the selected paragraphs.
	Line Height	Sets the line height of the selected paragraph.
	Insert an image	Inserts an image at the cursor location. Forms Designer stores the images in the SYSATTACHMENTS table.
	Insert a link toward an attached file	Inserts a link to an attachment at the cursor location. Forms Designer stores the attachments in the SYSATTACHMENTS table.
	Insert a link toward another record	Inserts a link to a record at the cursor location.
	Link	Inserts a hyperlink at the cursor location or edits the selected hyperlink. A user can jump to the associated anchor by clicking this hyperlink.
	Insert Table	Inserts a table at the cursor location. **Note:** Right-click the table to access the columns, rows, cells, and editing commands.
Ω	Insert Special Character	Inserts a specified character at the cursor location.
—	Insert Horizontal Line	Inserts a horizontal bar at the cursor location.
	Undo	Revert to last saved record.
	Redo	Redoes the reverted action.
	Code View	Toggles between source code mode and wysiwyg mode for editing.
/	Fullscreen	Switch between full screen and normal screen modes.

HTML Editor keyboard shortcuts

See Keyboard shortcuts for HTML Editor for detailed information.

Related topics

Form creation
Forms Designer
Web client forms
Building accessible forms
Access Forms Designer
Create a form using the Form Wizard
HTML Editor control
HTML Viewer control
Forms Designer controls and tools

1.1.1.1.6. Add a subform to a form

Applies to User roles: System Administrator

Subforms enable you to create a form once and then use it in other forms. You construct a subform in the same way as you would a standard form. You display a subform on another form by calling it with a subform widget embedded in the form.

Limitation: If a subform contains only a table, make sure that the table and subform have the same height. This way, a vertical scroll bar will appear for the table if needed when the total height of all table rows exceeds the specified table height. If you specify a table height greater than the subform height, the table rows may overflow outside the subform and overlap other fields on the form. If a subform contains multiple widgets, make sure that the subform height is set to be large enough to hold all the widgets; otherwise overlaps may also occur.

To add a subform to a form:

Follow these steps to add a subform to a form.

1. Click **Tailoring** > **Forms Designer** in the System Navigator, or type `fd` on the Service Manager command line and then press **Enter**.
2. Type a form name and click **Search**.
   Service Manager opens a list of forms. Click the form you want to update to open it.
3. Click **Design** to open design mode.
4. Add a subform widget, or select an existing subform.
   The properties for the comfill appear in the properties view.
5. Specify the database field to access in the Value text box of the Input property.
6. Specify the name of the form you want to display when the user hovers over the comfill in the Subform Format field.
7. Specify the virtual join field to associate with the comfill in the Subform Input field.
8. Mark the Subform Enabled checkbox.
9. Click **Web Preview** to see what the form will look like in the Web client.
10. Click **OK** to see what the form will look like in the Windows client.
11. Click **OK** to save your changes.
12. Open the form using Database Manager to test your changes. Click **Tailoring** > **Database Manager** and then click **Search**. Service Manager opens a list of forms. Click the form you just created to open it.

Sort the data in a subform

Applies to User roles: System Administrator

You can configure the default sort order of the data that's displayed in a subform. To do this, configure the following two drop-down lists in the Link form of the subform:

- **Sort Field(To/Target)**

Sets the field by which the subform is sorted

- **Sort Order(To/Target)**

Sets the order in which the field is sorted

Note

Note

- If multiple links are defined for a field in the Link file of the subform, you must use the first link to define the default sort field and sort order.
- You can't sort array fields in Service Manager 9.32. Therefore, you should not configure an array field as the default sort field.
- You may experience slow performance if you configure a long text field (for example, the **Description** field) as the default sort field.

Related topics

Form creation
Forms Designer
Using pop-ups
Using the drawing canvas
Access Forms Designer
Create a form using the Form Wizard
Update a form
Forms Designer controls and tools
Comfill control
Subform control

1.1.1.1.7. Using pop-ups

You can display a form as a pop-up by embedding the comfill widget in the form.
Pop-up forms enable you to create a form once, and then use it in other forms. Construct a pop-up form in the same way as you would a standard form.
Forms displayed as pop-ups are read-only; users can't interact with them. If you want your users to be able to interact with the form, display the form as a standalone of another form instead of as a pop-up.

- Pop-ups only display one record. If the virtual join returns zero or multiple records, the pop-up form will display an error message.
- Only a single pop-up can be active at a time.

The following widgets are supported for pop-ups. Use a regular subform instead of a pop-up subform for any widget not on this list. If you use a non-supported widget, the pop-up will display an error message.

- Checkbox
- Date
- Group/Frame
- Image
- Label
- Table
- Text
- TextArea
- Wrap Label

Note: For the Web client, the allowed maximum width of a pop-up form is 500 pixels. When designing a pop-up form for the Web client, make sure that the pop-up form is not too wide; otherwise, it may not appear correctly on the Web client.

Add a pop-up to a form

Applies to User roles: System Administrator

To add a pop-up to a form:

1. Click **Tailoring** > **Forms Designer** in the System Navigator, or type `fd` on the Service Manager command line and then press **Enter**.

2. Type a form name and click **Search**.
   Service Manager opens a list of forms. Click the form you want to update to open it.
3. Click **Design** to open design mode.
4. Add a comfill widget, or select an existing comfill.
   The properties for the comfill appear in the properties view.
5. Specify the name of the form you want to display when the user hovers over the comfill in the Pop-up Subform Format field.
6. Specify source field name to associate with the comfill in the pop-up Subform Input field.
   Note: This field should exist on the link record.
7. Click the **Pop-up Subform Enabled** box.
8. Click **Web Preview** to see what the form will look like in the Web client.
9. Click **OK** to see what the form will look like in the Windows client.
10. Click **OK** to save your changes.
11. Open the form using Database Manager to test your changes:
    1. Click **Tailoring** > **Database Manager**
    2. Click **Search**.
       Service Manager opens a list of forms.
    3. Click the form you just modified to open it.

When the user hovers over the comfill control that you placed on the form, if the virtual join returns a single record, Service Manager will display it in a pop-up subform.

Related topics

Form creation
Forms Designer
Using the drawing canvas
Access Forms Designer
Create a form using the Form Wizard
Add a subform to a form
Add a dynamic form to a form
Forms Designer controls and tools
Comfill control
Subform control

1.1.1.1.8. Add a dynamic form to a form

Applies to User Roles: System Administrator



Tip
Tip
For the format for the XML that describes the dynamic form, refer to `plugins\com.hp.ov.sm.client.common_<VERSION>\src\resources\schema\dynamicForm.xsd`.

To add a dynamic form to a form, follow these steps:

1. Click **Tailoring** > **Forms Designer** in the System Navigator, or type `fd` on the Service Manager command line and then press **Enter**.
2. Type a form name and click **Search**.
   Service Manager opens a list of forms. Click the form you want to update to open it.
3. Click **Design** to open design mode.
4. Drag the Dynamic form icon to the desired location.
5. Click the dynamic form field to select it.
6. Type the name of the form in the **Value** text box of the **Name** Property.
7. Type the name of the field in which to store the XML in the **Value** text box of the **Input** property

8. Specify the height, width, and other properties of the dynamic form, as desired.

   Note
   Note
   Forms designer sets the widget size set to the maximum size the dynamic form (once stretched) can occupy. You can resize the widget, but if you set the size too small, the dynamic form will appear truncated.

9. Click **Web Preview** to see what the form will look like in the Web client.
10. Click **Ok** to see what the form will look like in the Windows client.
11. Click **Ok** to save your changes.
12. To see the dynamic form, open the form on which it is located with Database Manager.
    Each time you save changes in Forms Designer, exit the form and reload it in Database Manager.

Best practice

To see the XML and the dynamic form at the same time for testing purposes, create a form with a regular text input field and a dynamic form widget for the same data field. Since the Web client doesn't allow you to use the same field name more than once on a form, alias the field, making the input property for the dynamic forms widget the alias and the input property for the text file the field itself. Be sure to make one of these read-only, so that they don't interfere with each other. Put your XML into the text field. You can use the magnify option to open a large text box to edit the XML.

Once you have created an acceptable dynamic form, you can generate the XML with a RAD or JavaScript application.

Related topics

Form creation
Forms Designer
Creating and editing forms
Using the drawing canvas
Forms Designer properties view
Setting properties
Forms Designer best practices
Building accessible forms
Access Forms Designer
Create a form using the Form Wizard
Update a form
Dynamic Form control
Forms Designer controls and tools

1.1.1.1.9. Building a graph diagram

The graph control enables you to display and interact with a graph that consists of a set of nodes connected by edges that represent a network of interrelated objects. The following is a summary of the diagram's capabilities:

- Supports hierarchical graphs. Since it's not feasible to draw all possible nodes on one diagram, the graph represents a subset of these as a single node. The user can then drill down to see the node or edge of interest.
- Supports inter-edges. An inter-edge is a virtual edge between two group nodes that represent the many edges that actually run between the two groups.
- Supports advanced layout algorithms that minimize edge crossover.
- Supports a client/server model. The application partitions between the client and the server.
- Supports the ability to bind compound rendering methods to nodes and edges. For example, a node may include an icon, two labels, and a circle around it.
- Supports user-definable actions on nodes and edges by double-clicking and/or right mouse click.
- Supports scaling.
- Supports scrolling and scaling.
- Supports printing and output in common file formats.

- Supports modifications to the diagram such as adding, moving, and deleting nodes and edges.

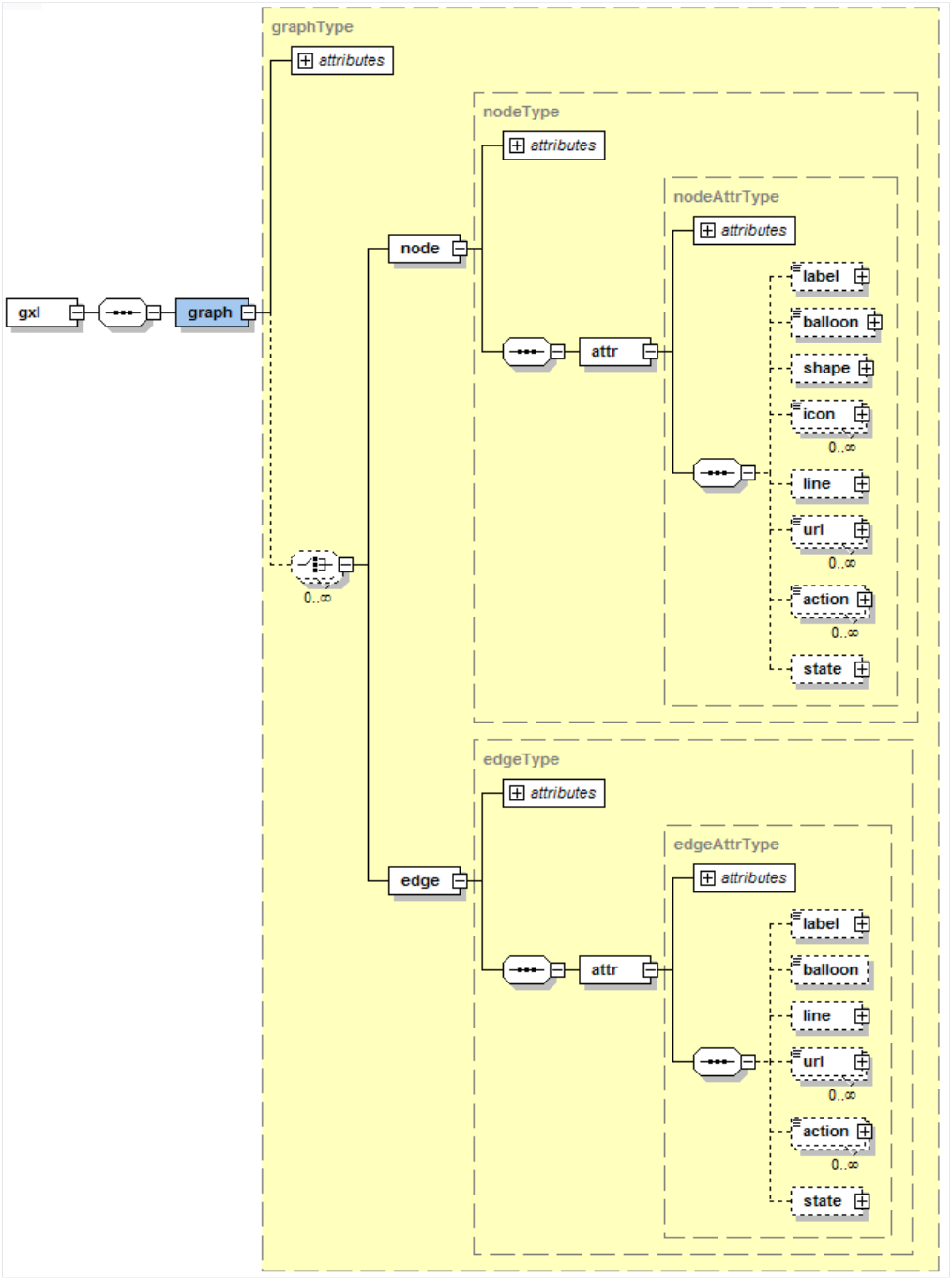
Components of the graph

The graph diagram provides a comprehensive visualization solution that enables the ability to easily visualize relationships between different components. An extensive library of graphics display the relationship in a variety of views. You can customize the views to fit your organization, including using your own graphics for display. You can view, add, or update relationships using the graphical interface. The main components of the graph diagram are:

- Graph
  A mathematical structure used to model relationships between objects from a certain collection.
- Node
  Representation of an entity. Nodes are defined by an ID and at a minimum require that either a bounding shape or an icon be defined so they are visible in the graph. Nodes can be joined by an edge (which also has an ID) by specifying the IDs of the joined nodes as the "to" and "from" attribute values of the node edge. Line styles and colors are optional.
- Edge
  Represents a relationship between two nodes. An edge is usually depicted as a line connecting its end nodes. An arrowhead at the target end of the line indicates the destination of a directed edge.

Defining the graph in XML format

The XML used to define a graph for Service Manager is an extension of a graph definition standard GXL. The GXL standard defines the basic structure for defining nodes and edges, but is lacking in specifics about graph visualization. The primary extension is the addition of an *attr* element for conveying visualization specific properties of the nodes and edges. The diagram below illustrates the primary elements involved in the definition. For more detail about the proper structure of the XML, see the link to the schema in the related topics.



W3C schema for graph diagram XML

This is the schema for validating a well formed graph definition. The schema is available as `dv/gxl-1.--dvd.xsd` in the war file.

```
<?xml version="1.0" encoding="UTF-8"?>
<!--W3C Schema generated by XML Spy v4.4 U (http://www.xmlspy.com)-->
<xs:schema xmlns="http://hp.com/schema/gxl-1.0-dv.xsd" xmlns:xs="
http://www.w3.org/2001/XMLSchema" targetNamespace="http://hp.com/schema/gxl-1.0-dv.xsd"
elementFormDefault="qualified">
   <xs:element name="gxl">
      <xs:complexType>
         <xs:sequence>
            <xs:element name="graph" type="graphType"/>
         </xs:sequence>
      </xs:complexType>
   </xs:element>
   <xs:complexType name="nodeAttrType">
      <xs:sequence>
         <xs:element name="label" type="labelType" minOccurs="0"/>
         <xs:element name="balloon" type="labelType" minOccurs="0"/>
         <xs:element name="shape" type="shapeType" minOccurs="0"/>
         <xs:element name="icon" type="iconType" minOccurs="0" maxOccurs="unbounded"/>
         <xs:element name="line" type="lineType" minOccurs="0"/>
         <xs:element name="url" type="urlType" minOccurs="0" maxOccurs="unbounded"/>
         <xs:element name="action" type="actionType" minOccurs="0" maxOccurs="unbounded"/>
         <xs:element name="state" type="nodeStateType" minOccurs="0"/>
      </xs:sequence>
      <xs:attribute name="name" type="xs:string" use="required" fixed="Viz"/>
   </xs:complexType>
   <xs:complexType name="edgeAttrType">
      <xs:sequence>
         <xs:element name="label" type="labelType" minOccurs="0"/>
         <xs:element name="balloon" type="xs:string" minOccurs="0"/>
         <xs:element name="line" type="edgeLineType" minOccurs="0"/>
         <xs:element name="url" type="urlType" minOccurs="0" maxOccurs="unbounded"/>
         <xs:element name="action" type="actionType" minOccurs="0" maxOccurs="unbounded"/>
         <xs:element name="state" type="stateType" minOccurs="0"/>
      </xs:sequence>
      <xs:attribute name="name" type="xs:string" use="required" fixed="Viz"/>
   </xs:complexType>
   <xs:complexType name="shapeType">
      <xs:attribute name="type" type="shapetypeEnum" use="required"/>
      <xs:attribute name="x" type="xs:short" use="optional"/>
      <xs:attribute name="y" type="xs:short" use="optional"/>
      <xs:attribute name="width" type="xs:short" use="optional"/>
      <xs:attribute name="height" type="xs:short" use="optional"/>
      <xs:attribute name="color" type="xs:string" use="optional"/>
   </xs:complexType>
   <xs:complexType name="edgeType">
      <xs:sequence>
         <xs:element name="attr" type="edgeAttrType"/>
      </xs:sequence>
      <xs:attribute name="id" type="xs:ID" use="required"/>
      <xs:attribute name="from" type="xs:string" use="required"/>
      <xs:attribute name="to" type="xs:string" use="required"/>
   </xs:complexType>
   <xs:complexType name="graphType">
      <xs:choice minOccurs="0" maxOccurs="unbounded">
         <xs:element name="node" type="nodeType"/>
         <xs:element name="edge" type="edgeType"/>
      </xs:choice>
      <xs:attribute name="id" type="xs:NMTOKEN" use="required"/>
   </xs:complexType>
   <xs:complexType name="iconType" mixed="true">
      <xs:sequence>
         <xs:element name="path" type="xs:string" minOccurs="0"/>
         <xs:element name="label" type="labelType" minOccurs="0"/>
         <xs:element name="balloon" type="labelType" minOccurs="0"/>
         <xs:element name="url" type="urlType" minOccurs="0" maxOccurs="unbounded"/>
      </xs:sequence>
      <xs:attribute name="group" type="xs:string" use="optional"/>
      <xs:attribute name="position" type="positionType" use="optional"/>
   </xs:complexType>
   <xs:complexType name="labelType">
      <xs:simpleContent>
         <xs:extension base="xs:string">
            <xs:attribute name="font" type="xs:string" use="optional"/>
            <xs:attribute name="style" type="xs:short" use="optional"/>
            <xs:attribute name="size" type="xs:short" use="optional"/>
            <xs:attribute name="width" type="xs:short" use="optional"/>
         </xs:extension>
      </xs:simpleContent>
   </xs:complexType>
```

```
<xs:complexType name="lineType">
    <xs:attribute name="color" type="xs:string" use="optional"/>
    <xs:attribute name="style" type="lineStyleType" use="optional"/>
    <xs:attribute name="weight" type="xs:short" use="optional"/>
    <xs:attribute name="group" type="xs:string" use="optional"/>
</xs:complexType>
<xs:complexType name="edgeLineType">
    <xs:complexContent>
        <xs:extension base="lineType">
            <xs:attribute name="end" type="lineEndType" use="optional"/>
            <xs:attribute name="highlightcolor" type="xs:string" use="optional"/>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:complexType name="stateType">
    <xs:attribute name="hidden" type="xs:boolean" use="required"/>
</xs:complexType>
<xs:complexType name="nodeStateType">
    <xs:complexContent>
        <xs:extension base="stateType">
            <xs:attribute name="expanded" type="xs:boolean" use="required"/>
            <xs:attribute name="opened" type="xs:boolean" use="required"/>
            <xs:attribute name="root" type="xs:boolean" use="optional"/>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:simpleType name="lineStyleType">
    <xs:restriction base="xs:short">
        <xs:enumeration value="0"/>
        <xs:enumeration value="1"/>
        <xs:enumeration value="2"/>
        <xs:enumeration value="3"/>
        <xs:enumeration value="4"/>
        <xs:enumeration value="5"/>
    </xs:restriction>
</xs:simpleType>
<xs:simpleType name="lineEndType">
    <xs:restriction base="xs:short">
        <xs:enumeration value="0"/>
        <xs:enumeration value="1"/>
        <xs:enumeration value="2"/>
        <xs:enumeration value="3"/>
    </xs:restriction>
</xs:simpleType>
<xs:simpleType name="shapetypeEnum">
    <xs:restriction base="xs:short">
        <xs:enumeration value="-1"/>
        <xs:enumeration value="0"/>
        <xs:enumeration value="1"/>
        <xs:enumeration value="2"/>
        <xs:enumeration value="3"/>
    </xs:restriction>
</xs:simpleType>
<xs:complexType name="nodeType">
    <xs:sequence>
        <xs:element name="attr" type="nodeAttrType"/>
    </xs:sequence>
    <xs:attribute name="id" type="xs:NMTOKEN" use="required"/>
</xs:complexType>
<xs:complexType name="urlType">
    <xs:simpleContent>
        <xs:extension base="xs:string">
            <xs:attribute name="urltype" type="urltypeValue" use="required"/>
            <xs:attribute name="label" type="xs:string"/>
            <xs:attribute name="target" type="xs:string"/>
        </xs:extension>
    </xs:simpleContent>
</xs:complexType>
<xs:complexType name="actionType">
    <xs:simpleContent>
        <xs:extension base="xs:string">
            <xs:attribute name="type" type="xs:string" use="required"/>
            <xs:attribute name="label" type="xs:string"/>
            <xs:attribute name="balloon" type="xs:string"/>
            <xs:attribute name="forcemenu" type="xs:boolean"/>
            <xs:attribute name="target" type="xs:string"/>
        </xs:extension>
    </xs:simpleContent>
</xs:complexType>
```

```
  <xs:simpleType name="urltypeValue">
    <xs:restriction base="xs:string">
      <xs:enumeration value="HTML"/>
      <xs:enumeration value="XML"/>
      <xs:enumeration value="XMLADD"/>
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="positionType">
    <xs:restriction base="xs:string">
      <xs:enumeration value="C"/>
      <xs:enumeration value="N"/>
      <xs:enumeration value="NE"/>
      <xs:enumeration value="E"/>
      <xs:enumeration value="SE"/>
      <xs:enumeration value="S"/>
      <xs:enumeration value="SW"/>
      <xs:enumeration value="W"/>
      <xs:enumeration value="NW"/>
    </xs:restriction>
  </xs:simpleType>
</xs:schema>
```

Valid values for graph elements and attributes

These are the valid values for the elements and attributes.

Element/attribute name	Type
action	String specifying an action acted on by the associated action handler.
action/balloon	String specifying tooltip to display if action is shown in a menu.
action/forcemenu	If set to true, indicates the pop-up should be display when only one labeled action is defined.
action/label	String specifying a label to display if action is shown in a menu.
action/type	String specifying the type of action invoked by the action handler.
balloon	String specifying the balloon text to show for the item.
edge/from	String specifying the ID of the originating node.
edge/id	String specifying a unique ID for the edge.
edge/to	String specifying the ID of the target node.
icon	String specifying relative or absolute URL or relative resource path.
icon/group	String specifying a display group name.
icon/position	String specifying icon position from one of the possible values relative to the node center. The default is C. N = North or above NE = Northeast or upper right E = East or right SE = Southeast or lower right S = South or below SW = Southwest or lower left W = West or left NW = Northwest or upper left C = Center
label/font	String specifying font name per Java Font class.
label/style	Integer specifying font style per Java Font class.
label/size	Integer specifying font size per Java Font class.
line/color	String specifying color name or Hex RGB color value prefixed with #. The default is black.
line/end	Integer specifying the type of line end. The default is 0. NO ARROW = 0 SOURCE_ARROW = 1 TARGET_ARROW = 2 SOURCE_AND_TARGET_ARROW = 3

line/highlightcolor	String specifying highlight color name or Hex RGB color value prefixed with #. The default is none.
line/style	Integer specifying line style. The default is 0. SOLID = 0 LONG_DASH = 1 SHORT_DASH = 2 DOT = 3 DASH_DOT = 4 DASH_DOT_DOT = 5
line/weight	Integer specifying the line weight. The default is 2.
node/id	String specifying a unique id for the node.
shape/color	String specifying fill-color name or Hex RGB color value prefixed with #. The default is none.
shape/height	Integer specifying height.
shape/type	Integer specifying the type of shape. The default is -1. NONE= -1 RECTANGLE = 0 ELLIPSE = 1 DIAMOND = 2 ROUNDED_RECT = 3
shape/width	Integer specifying width.
shape/x	Integer specifying x coordinate.
shape/y	Integer specifying y coordinate.
state/expanded	If set to true, indicates the node is currently expanded.
state/hidden	If set to true, indicates the node or edge is currently hidden.
state/opened	If set to true, indicates the node has been opened by reading the associated XMLADD action or URL.
state/root	If set to true, indicates the node is the root node for the graph.
url	String specifying a relative or absolute URL path
url/balloon	String specifying tooltip to display if URL is shown in a menu.
url/label	String specifying a label to display if URL is shown in a menu.
url/urltype	String specifying the type of URL. The default is HTML. HTML = new HTML page XML = new XML defining graph XML_ADD = XML content to add to current graph

Referencing images in the XML for the graph control

When you are building an application that includes a graph diagram, you can reference an image that's attached to a record in Service Manager. The graph uses the attached image to display the icon for that node. You can also display portrait icons that are attached to contact records in the graph diagram.

This is an example of a schema definition for an icon.

```
<xs:complexType name="iconType" mixed="true">
  <xs:sequence>
    <xs:element name="path" type="xs:string" minOccurs="0"/>
    <xs:element name="label" type="labelType" minOccurs="0"/>
    <xs:element name="balloon" type="labelType" minOccurs="0"/>
    <xs:element name="url" type="urlType" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:attribute name="group" type="xs:string" use="optional"/>
  <xs:attribute name="position" type="positionType" use="optional"/>
</xs:complexType>
```

The syntax to reference an image from the database and display it in the graph is:

`scattach://{type}:{uid}:{attachmentname}:{table}:{record}`, where *type* equals `img` or `file`.

The unique identifier for each image is located on the corresponding SYSATTACHMENTS record.

The following is an example of how to reference an image that's an attachment to a record.

`scattach://file:45c0dfd600245044034b76c8:max_manager_small.gif:contacts:MANAGER, MAX`

The following is an example of how to reference a portrait image on a contacts record.

`scattach://img:45afb84a002d42c403030d48:MANAGER, MAX:contacts:MANAGER, MAX`

Here is an example of a complete and valid XML statement that references an image. This particular statement references an image that's attached to Max Manager's contacts record. This displays as an icon for the node in the graph diagram.

```
<?xml version="1.0" encoding="UTF-8"?>
<gxl>
  <graph id=\"\">
    <node id=\"MANAGER, MAX\">
      <attr name=\"Viz\">
        <label font=\"Times New Roman\" size=\"12\" style=\"1\">MANAGER, MAX</label>
        <balloon>Max Manager</balloon>
        <icon>scattach://file:45c0dfd600245044034b76c8:max_manager_small.gif:contacts:MANAGER, MAX</icon>
        <action label=\"Expand\" type=\"XMLADD\">XMLADD</action>
        <action label=\"Action 1\" type=\"Action 1\">Action 1</action>
        <action label=\"Action 2\" type=\"myaction2\">Action 2</action>
      </attr>
    </node>
  </graph>
</gxl>
```

Icons with larger position values appear on top of smaller position values unless the images specify the same value for the position attribute. In that case, the icons are rendered in document order so that icons later in the document for the same node overlay those preceding them. The origin for all icons is the center. When you specify a group attribute for an icon, it includes items that contain duplicate group attributes.

Defining actions on nodes and edges in the XML for the graph control

When you are building an application that includes a graph diagram, you can define the actions on nodes and edges.

Following is the pertinent part of the schema definition for an action on a node or an edge. The *type* attribute is required and can be any value if a custom action handler is provided. If the default action handler is used, then the possible values are the same as the *urltype* attribute for the *url* element. If a *label* attribute is present, the action appears on the popup action menu when more than one labeled action is available. Providing a value of true with the *forcemenu* attribute causes the popup to appear even when there is only a single possible action.

```
<xs:complexType name="actionType">
  <xs:simpleContent>
    <xs:extension base="xs:string">
      <xs:attribute name="type" type="xs:string" use="required"/>
      <xs:attribute name="label" type="xs:string"/>
      <xs:attribute name="balloon" type="xs:string"/>
      <xs:attribute name="forcemenu" type="xs:boolean"/>
      <xs:attribute name="target" type="xs:string"/>
    </xs:extension>
  </xs:simpleContent>
</xs:complexType>
```

The following is an example of a complete and valid XML statement that defines an expand action on the first node and two additional actions on each node and edge. It also references the images that are attached to the contacts records of Max Manager and Steve Supervisor. These display as icons for each node in the graph diagram.

```
<?xml version="1.0" encoding="UTF-8"?>
<gxl>
  <graph id=\"\">
    <node id=\"MANAGER, MAX\">
      <attr name=\"Viz\">
        <label font=\"Times New Roman\" size=\"12\" style=\"1\">MANAGER, MAX</label>
        <balloon>Max Manager</balloon>
        <icon>scattach://file:45c0dfd600245044034b76c8:max_manager_small.gif:contacts:MANAGER, MAX</icon>
        <action label=\"Expand\" type=\"XMLADD\">XMLADD</action>
        <action label=\"Action 1\" type=\"Action 1\">Action 1</action>
        <action label=\"Action 2\" type=\"myaction2\">Action 2</action>
      </attr>
    </node>
    <node id=\"SUPERVISOR, STEVE\">
      <attr name=\"Viz\">
        <label font=\"Times New Roman\" size=\"12\" style=\"1\">SUPERVISOR, STEVE</label>
        <balloon>Steve Supervisor</balloon>
        <icon>scattach://file:45c10d0b00116014030be8e0:steve_supervisor.gif:contacts:SUPERVISOR, STEVE</icon>
        <action label=\"Action 1\" type=\"Action 1\">Action 1</action>
        <action label=\"Action 2\" type=\"myaction2\">Action 2</action>
      </attr>
    </node>
    <edge from=\"SUPERVISOR, STEVE\" id=\"SUPERVISOR, STEVE - MANAGER, MAX\" to=\"MANAGER, MAX\">
      <attr name=\"Viz\">
        <line color=\"#000000\" style=\"0\" weight=\"2\"/>
        <label font=\"Times New Roman\" size=\"12\" style=\"1\">Manages</label>
        <action label=\"Edge Action 1\" type=\"Edge Action 1\">Edge Action 1</action>
        <action label=\"Edge Action 2\" type=\"myaction2\">Edge Action 2</action>
      </attr>
    </edge>
  </graph>
</gxl>
```

Graph toolbar icons

The following table describes the icons available for use on the graph diagram:

Name	Icon	Description
Zoom In		Decreases the size of the current visualization diagram.
Zoom Out		Enlarges the size of the current visualization diagram.
Fit Content		Fits the content of the diagram to the view.
Select Mode		Highlights an item in the visualization diagram.
Window Area Mode		Clicking and dragging the cursor to highlight one or more items will zoom in and fit the selection to the current view.
View Drag Mode		Clicking the mouse and dragging the cursor will pan the diagram within the current viewable area.
Drill Down Mode		Drills down further into a node and its relationships.
Show Overview		Turns on the overview window displaying an overview of the entire graph contents highlighting what's displayed in the working window.
Show Node Labels		Shows or hides the labels on nodes.
Show Edge Labels		Shows or hides the labels on edges.
Print Preview		Displays a print preview of the visualization diagram in the current focus. You can then select Print from the Print Preview dialog.
Select Layout		Filter the view of the graph using one of the available layouts.

Functions and displayevent definitions for actions in the graph diagram

These functions enable users to interact with a graph that represents a network of interrelated objects. Both RAD and JavaScript functions are supported.

Graph event

The following are the functions and displayevent definition that are required to get and display more information in a graph diagram. When a user requests more information in the graph diagram, the applications generate additional XML that's routed through the server to the client for re-rendering.

Displayevent
    OnGraphEvent (32105)
RAD functions

get.graph.id()
get.graphnode.id()
add.graphnodes(xml)

JavaScript functions
get_graph_id()
get_graphnode_id()
add_graphnodes(String strXML)

Graph action

The following are the functions and displayevent definition that are required to activate an action that's defined for a node. When a user selects an action from the pop-up menu for a node, the application can access the Graph ID, Action, and Target by using the functions below. This information can be used in the displayevent to call additional application logic.

Displayevent
OnGraphAction (32100)
RAD functions
get.graph.action()
get.graph.target()

JavaScript functions
get_graph_action()
get_graph_target()

Activate an action on a node or edge in a graph

Applies to User Roles: System Administrator

To activate an action on a node or edge in a graph:

1. Click once on a node or edge inside of a graph diagram.
   A pop-up menu opens with a list of available actions.
2. Click on a node or edge action from the pop-up menu.
   The action and target run.

Note: Right-click or left-click any of the toolbar controls to activate an action.

Set the focus on the graph

Applies to User Roles: System Administrator

To set the focus on the graph:

1. Type `fd` on the Service Manager command line and press Enter to start Forms Designer.
2. Type a form name and click **Search** or create a new form.
3. Once in design mode, click the **Graph** control in the form.
   This highlights the Graph control and enables you to edit its properties.
4. In the Properties view, set the Input property value to a database field or variable. For example, `$L.graph`.
   This sets the focus on the graph and enables the end user to perform an action on the graph (such as view a node detail) and then return to the tab with the graph.

Related topics

Graph control

1.1.1.2. Forms Designer controls and tools

Forms Designer has the following controls and tools. They are listed in alphabetical order and each control has different properties. Click the Name link to view the details.

Control or Tool	Name	Description
	Activity Viewer control	Adds an Activity Viewer that enables users to view activity logs.
	Attachments control	Adds a box into which users can place non-Service Manager documents (for example, from Microsoft Word or Microsoft Excel).
	Button control	Adds a button that activates a Control ID when clicked. You can use text, graphics, or both to customize buttons.
	Calendar control	Adds a calendar.
	Chart control	Adds a chart that displays the contents of a numeric array as a two-dimensional, color-coded bar chart with optional, definable buttons.
	Checkbox control	Adds a checkbox that displays and enables users to update the contents of a Boolean (logical) field, which can evaluate to true, false, unknown, or null.
	CMS Topology control	Displays the Topology and Impact through the UCMDB Gateway.
	Combo Box control	Adds a Combo Box that enables users to click a button and select from a drop-down list.
	Comfill control	Adds a combination Combo Box and fill button. You can specify a virtual join subform to display in a pop-up when the user hovers over this field.
	Convert Form Layout tool	Converts all of the fields on the left side of the form into one group, and converts each of the notebook tabs on the right into a group, arranged vertically, under the left side fields. In the Web client, these groups are collapsible.
	Date control	Adds a text box that displays and enables users to update the contents of a date field.

	Decimal control	Adds an input field that displays and enables users to update the contents of a numeric field and offers spinner buttons to increase or decrease a value.
	Dynamic Form control	Adds a dynamic form that becomes visible when populated by XML code from a RAD application or JavaScript.
	Embedded Viewer control	Adds a viewer that enables Web client users to view integrations.
	File control	Adds an Open File dialog box.
	Frame control	Adds a container that groups logically associated items.
	Graph control	Displays a diagram of connected nodes representing a system of interrelated things.
	Grid view tool	Toggles the alignment grid that you can use while designing the form.
	Group control	Adds a container that groups logically associated items and includes a text label at the top.
	HTML Editor control	Adds an HTML Editor that enables users to add standard HTML to the form.
	HTML Viewer control	Adds an HTML Viewer that enables users to view the HTML created using the HTML Editor.
	Image control	Adds a place to display a Microsoft Windows image file.
	Journal Updates Viewer control	Adds a Journal Updates Viewer that enables users to view journal updates.
	Label control	Adds a place for you to type a title for the form, a label for an object within the form, or otherwise place a single line of text on the form.
	Link label control	Adds a label that includes a caption and an icon that links to another source.
	List Builder control	Adds a compound widget that allows the user to construct a list of choices.

	Marquee control	Adds an attention-getting text message that scrolls continuously across the screen from right to left.
	Notebook control	Adds a container that subdivides the contents of a screen into logical groups or categories.
	Notebook Tab control	Adds a container for a notebook that has two or more pages. Navigate to each page by selecting its tab.
	Radio button control	Adds a radio button that enables users to select one value from a set of mutually exclusive values.
	Script control	Adds a place for users to enter JavaScript into a color-coded syntax editor.
	Selection tool	Enables you to select, deselect, and modify design objects.
	Subform control	Adds a sub-form to a larger form.
	Table column control	Adds a column to an existing table.
	Table control	Adds a table that displays one or more columns of data in a scrollable pane.
	Tag control	Adds a text box that displays the contents of a text field and conditionally enables users to enter one or more tags.
	Text area control	Adds a text area that displays the contents of a text field and conditionally enables users to input several lines of data.
	Text control	Adds a text box that displays the contents of a text field and conditionally enables users to enter or modify its contents.
	Timer control	Adds a clock timer.
	Web Preview tool	Enables you to see what the form will look like when displayed by the Web client.
	Wrap label control	Adds a multiline label.

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Add a subform to a form

1.1.1.2.1. Activity Viewer control

Use this control to add an Activity Viewer that enables users to view the activity logs.

To place an Activity Viewer on a form, click **Activity Viewer** and then click the form.

Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button, or checkbox, you will almost always use the default of 0.
Accessible Description	(Optional) Specify a component description for use with accessibility software. **Note:** This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Update Type Value List	Use this property in conjunction with the **Input** property to define journal update types that are identified in the database. Enter values by using the Edit List dialog box. You can either enter hardcoded entries for each list or supply a variable as the first and only entry. The system uses run-time values of the variable to populate these lists.

Login Name Conversion	Select this option to display the operator's Full Name for activity logs. If not selected, the operator's name will be displayed according to the environment settings.

1.1.1.2.2. Attachments control

Use this control to add a box into which users can place non- Service Manager documents (for example, from Microsoft Word or Microsoft Excel).

To place an attachment container on a form, click **Attachments** and then click the form.

Note: To save attachments with incident records using Qopen (Quick Open), the Delay Assigning Problem Number field in the Incident Management Environment profile must be cleared. If Delay Assigning Problem Number is selected (set to true), then attachments aren't saved with the incident record when performing a Qopen. Only the Qopen operation is affected by this setting.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button, or checkbox, you will almost always use the default of 0.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.
Read-Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to true.
Maximum Attachment Size	Specify the absolute size of an attachment, in bytes. Size limits defined in the Maximum Attachment Size field in a user's operator record override the size limit specified in the Company record. A non-zero value overrides any values specified in the Company or Operator record.

Maximum Attachment Size Condition	Specify a RAD expression to determine the maximum allowed size of an attachment based on some condition such as the capabilities of a user. For example, you can write a RAD expression that checks an operator's capabilities word and assigns a system administrator a higher attachment size limit than a manager, who in turn could have a higher attachment size limit than a service desk user.
Total Attachment Size	Specify the maximum amount of memory, in bytes, that all attachments in a form can use. The total size of all attachments must be lower than this threshold or the form no longer accepts additional attachments.
Total Attachment Size Condition	Specify a RAD expression to determine the total amount of memory that all attachments in the form can use based on some condition such as the capabilities of a user.
Maximum Attachments Allowed	Select whether you want to allow one or any number of attachments.

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1.1.1.2.3. Button control

Use this control to add an input field that displays and enables users to update the contents of a numeric field and offers spinner buttons to increase or decrease a value.

To place a button on a form, click **Button** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.



Note

Do not use the value 888 as a Button ID. The value 888 is a reserved number.

Best practices

- Use easy to understand names and descriptions for buttons.
- Make all buttons on a form the same type (all image, all text button or all text link).
- Place navigational buttons along the left edge or the top border of the form.
- Place other buttons at the top or bottom border.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Justification	Specify the justification for the object label as Left, Right, or Center.
Button ID	Specify a numeric identification that specifies a Control ID to transmit when clicked.
Button ID Condition	Specify an expression to override the Button ID property when that expression evaluates to true.

Image File	Specifies the name of an image file to display on the control. The image file must reside in any of the following paths, in which Service Manager looks for the image: **Windows client**: • <Service Manager>\Client\plugins\com.hp.ov.sm.client.common_x.xx.xxx\src\resources\icons • A local directory on the client machine, which is specified in the **Image path** field (**Window** > **Preferences**). For example: C:\smCustomImages. **Web client**: • <Service Manager web tier>\images. For example: <Tomcat>\webapps\webtier-9.61\images. **Note** When you specify the image file for this property, you need to provide both the file name and the file name extension.
Image Condition	Specify an expression to override the Image property when that expression evaluates to true.
Balloon Help	Specify a text string that appears when the cursor is held over the button to offer more information about the button's function.
Button Action	Specify an action for the button.

Button Action Params	Specify a parameter for the button action. **Note** The **OpenSmartSearchWidget** button action doesn't need a parameter, which opens smart search windows if smart search is configured. For the other button actions: **OpenDashboardTab**: Try to open `/dashboard.do?id=actionParams`. Action Params should be dashboardId. If no dashboard found, a new null dashboard will be opened. **OpenRuleSetEditor**: Try to open `/workflow/fileRuleSetEditor.jsp?tableName=actionParams`. Action Params should be tableName. If no object found, a new null object RuleSet editor will be opened. **OpenWorkflowORFileRuleSetEditor**: Try to open `/workflow/workflowEditor.jsp?actionParams`. If Button Action Params is set to `fileRuleSetEditor*`, then try to open `/workflow/fileRuleSetEditor.jsp?actionParams`.
Enabled	Specify an initial state.
Enabled Condition	Specify a condition that enables the button when a table row is selected. Example: [%tablename] == "selected" or [%tablename] != "selected" where tablename is the value of the Name property of the table on the screen.

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1.1.1.2.4. Calendar control

Use this control in Forms Designer to add a calendar to a form. To place a calendar on a form, click **Calendar** and then click the form.



Note

Note

The calendar widget is displayed well only in the web client. When you view the form from the Windows client, this widget is displayed as a blank area.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object. **Example**: `calendar`
X	Specify the object's horizontal position based on the left edge of the object. **Example**: 0
Y	Specify the object's vertical position based on the top edge of the object. **Example**: 1
Width	Specify the width of the object in alignment grid units. **Example**: 223
Height	Specify the height of the object in alignment grid units. **Example**: 44
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form. **Example**: Select this option.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true. **Example**: (empty)
Input	Specify the database field or variable to associate with this control. The primary use for this property is to set the focus on the graph. **Example**: (empty)
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button, or checkbox, you will almost always use the default of 0. **Example**: 0

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note** Note This function is only supported by the web client. If no description is present, the web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the web client will use the component's text. Accessibility software can query the web client to get this information. The client uses that information to present data to the user in a variety of ways. For example, speech simulation software will say the name and the type of component when the component gets focus, based on the settings in the Accessible Name and Accessible Description fields. **Example**: (empty)
Date	Specify the start date for data display in the calendar. Currently only the `$start.date` value is supported. This variable takes the value of the field specified in **Field for Initial Date** of each Embedded Calendar Configuration setting of an object (Change, Incident, and so on). See Configure search filters for the full calendar. For example, if you specify the **Planned Start** field as the **Field for Initial Date** for the Change object, in a change record whose Planned Start is May 23, 2013, the calendar displays data only for dates no earlier than May 23, 2013.
Predefined Query	Specify a query for data display in the calendar. Currently only the `$predefined.query.string` value is supported. This variable takes the query specified in each Embedded Calendar Configuration setting of an object. See Configure search filters for the full calendar.
Supported Views	Specify the views that you want the calendar to support. The value can be any combinations of `day`, `week`, and `month`, separated with a semicolon (;). If you leave this property empty, all three views are supported. **Example**: `day;week`

Default View	Specify the initial view when the calendar is opened. Currently only the `$default.view` value is supported. This variable takes the Default View value specified in each Embedded Calendar Configuration setting of an object. See Configure search filters for the full calendar. For example, if you specify Week as the Default View of the embedded calender configuration for the Change object, the calendar is displayed in the weekly view in the **Calendar** section of each Change record.
View Type	Specify the view type that you want the calendar to support. Select **Calendar** to display a normal view, or select **SLM** to display the SLT view. If you leave this property empty, the system displays a normal calendar view.

1.1.1.2.5. Chart control

Use this control to add a chart that displays the contents of a numeric array as a two–dimensional, color–coded bar chart with optional, definable buttons.

To place a chart on a form, click **Chart** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a list of values to display in the chart. Example: 3;4;10;3
Input	Specify the database field or variable to associate with this control. For more information on what to pass to the field or variable refer to the Publish and Subscribe Help.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Foreground Color	Select the text color from the drop-down list.
Balloon Help	Specify the text string to display when the cursor is held over the button to offer more information about the button's function.
Bar Width	Specify the width of bars in alignment grid units.
Scale Max	Specify the maximum vertical bar height. Columns that exceed the Scale Max setting appear clipped. A setting of 0 causes the chart to automatically scale vertically to the tallest column.
Button Base	Select this option to enable a row of buttons across the base of the chart, one button per column.

Base Button ID	Specify the Control ID for the left-most chart button. It is incremented by one from left to right. Using the default value of 801 as an example, the left-most button would transmit 801, the second button from the left would transmit 802, the third 803, and so forth. The Base Button ID relates to an option number in the menu record. Assign Button IDs to the chart and create an associated option number in the menu record, so that each button calls a stored query.
Color List	Specify a list of semicolon delimited colors. The Color Scale and Color Percent properties determine how the list is applied to the columns. The available colors are: black, red, green, blue, gray, light gray, dark gray, yellow, cyan, magenta, white, forest, navy, purple, teal, brick, and manila.
Color Scale	Specify a list of semicolon delimited numeric values to determine the color of a column.
Color Percent	Select this option to use the Color Scale property a list of percentages from 1 to 100.

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1.1.1.2.6. Checkbox control

Use this control to add a checkbox that displays and enables users to update the contents of a Boolean (logical) field, which can evaluate to true, false, unknown, or null.

To place a checkbox on a form, click **Checkbox** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Accessible Name	Specify a component name for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no name is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, it uses the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, speech simulation software says the name and the type of component when the component has focus.

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information. The client uses that information to present data to the user in a variety of ways. For example, speech simulation software will say the name and the type of component when the component gets focus, based on the settings in the Accessible Name and Accessible Description fields.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Foreground Color	Select the text color from the drop-down list.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.
Read-Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to true.
Array Length	Specify the size of the scrolling region used to view array entries. A scroll bar appears beside the fields to allow users to view the array entries. - If a field is assigned an Array Length of 5, the form stacks five fields vertically to allow users to view the five array entries. - If Array Length is set to -1, the form dynamically creates as many fields as there are entries in the array. - If the Input data type is scalar, only a single text box appears. **Note**: The screen object must be associated with an array data structure. The default is 0, which means one vertical line of information appears.
Data Changed Event	Specify the option number (such as Button ID) to call if the data contained in the object changed. This sends an event to the display RAD application.

1.1.1.2.7. CMS Topology control

Use this control to add a CMS Topology that enables Web client users to view the Topology and Impact through the UCMDB Gateway.

To place a CMS Topology on a form, click **CMS Topology** and then click the form.

Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Input	Specify the database field or variable to associate with the global ID. Only the value corresponding to the global ID in UCMDB is valid.
Customer Id	Specify the customer ID.
Type	Specify the type of the CMS Topology. Select **Impact** to view the Show Impact widget, or select **Topology** to view the Topology widget.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button, or checkbox, you will almost always use the default of 0.
Accessible Description	(Optional) Specify a component description for use with accessibility software. **Note:** This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.

1.1.1.2.8. Combo Box control

Use this control to add a Combo Box that enables users to click a button and select from a drop-down list. The items in the list are associated with a database field or variable. Users can also type in a value if the checkbox in the selectonly property is cleared (set to false).

To place a Combo Box on a form, click **Combo Box** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Best practice[edit]

Use eight drop–down lines unless the drop–down set contains a fixed number of elements less than or equal to seven, in which case the number of lines should be equal to the number of elements in the drop–down set.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Accessible Name	Specify a component name for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no name is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, it uses the component's text. Accessibility software can query the Web client to get this information and use it to present data to the user in a variety of ways. For example, speech simulation software says the name and the type of component when the component has focus.

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.
Read-Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to true.
Mandatory	Select this option so that a red asterisk indicating a required field is displayed. This is a visible change only. - To make the field mandatory for any form it appears on, use the System Definition Utility. - To make the field mandatory for a small number of forms only, use Format Control.
Mandatory Condition	Specify an expression to override the Mandatory property when that expression evaluates to true.
Mandatory For Read-Only	If you set this option to true, the system displays the mandatory icon on the read-only Combo Box widgets. You must add appropriate accessible descriptions to guide users how to fill the value for the mandatory and read-only fields. **Note** Note This function is only supported by the Web client.

Array Length	Specify the size of the scrolling region used to view array entries. A scroll bar appears beside the fields to allow users to view the array entries. - If a field is assigned an Array Length of 5, the form stacks five fields vertically to allow users to view the five array entries. - If Array Length is set to -1, the form dynamically creates as many fields as there are entries in the array. - If the Input data type is scalar, only a single text box appears. **Note**: The screen object must be associated with an array data structure. The default is 0, which means one vertical line of information appears.
Password	Select this option to make the text entered in the field a series of asterisks on the screen. The data is not encrypted, just transposed.
Maximum Chars	Specify the number of characters the user can enter in the field. The default is 0 (unlimited).
Maximum Characters Beep	Select this option to notify the user by sounding a beep that they reached the maximum number of characters.
Case Conversion	Specify how to change the case of text entered in the field. The choices are None, Upper, Lower.
Decimals	Specify the number of decimal points for numbers in this field. Service Manager modifies numbers entered in the field to the number of decimal points you specify.
Parse	Select this option to verify that the text entered in a field is the correct syntax. The syntax for the field is defined by the type of field it displays (date, expression, number, and so forth).
Data Changed Event	Specify the option number (such as Button ID) to call if the data contained in the object changed. This sends an event to the display RAD application.

Value List	In conjunction with the Input property, defines how the Display List values are identified in the database. Value and Display Lists are entered using the Edit List dialog box. You can enter hard coded entries for each list, or you can supply a variable as the first and only entry. The run time values of the variable are used to populate these lists. This property is supported only in association with the toolbar configuration Knowledge Management Extended, which is not intended to be used outside of the out-of-box Knowledge Management module. **Note**: When defining Value List and Display List properties, you should avoid using the following reserved characters: • semicolon (;) • tab key • newline (carriage return) • form feed • backspace • equal sign (=) **Important**: Do not use keys (like backspace), as the system can't process their ASCII representation entries correctly which causes unpredictable results.
Value List Condition	Specify an expression to override the property when that expression evaluates to true.

Display List	Specify the values that appear in the drop-down list at run time. There must be a one–to–one correspondence between the values for Value List and for Display List. If the Display List is the only populated property, the display values are written to the database. This property is supported only in association with the toolbar configuration Knowledge Management Extended, which is not intended to be used outside of the out-of-box Knowledge Management module. **Note**: When defining Value List and Display List properties, you should avoid using the following reserved characters: • semicolon (;) • tab key • newline (carriage return) • form feed • backspace • equal sign (=) **Important**: Do not use keys (like backspace), as the system can't process their ASCII representation entries correctly which causes unpredictable results.
Display List Condition	Specify an expression to override the Display List property, when the expression evaluates to True.
Box Lines	Specify the window size of the drop-down list.
Select Only	Select this option to require the user to select from the Combo Box drop-down list. If the user makes a manual entry, the value must already exist in the drop-down list (defined in the Value List and Display List properties). Clear this option to allow the user to select from the drop-down list or to manually enter a value.

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1.1.1.2.9. Comfill control

Use this control to add a combination Combo Box and Fill button. Comfill has all the properties of a Combo Box, plus the capability for Fill button. You can select which comfill buttons to display. For example, to display just a Fill button, set the Third Button Visible property to true by selecting the checkbox and set the Combo Button Visible property to false, by clearing the checkbox.

You can specify a form to display as a pop-up when the user hovers over the comfill. Subforms displayed as pop-ups are read-only; users can't interact with them. If you want your users to be able to interact with the form, you must display the subform on another form instead of as a pop-up. To display the form as a subform, use the subform control.

To place a comfill box on a form, click **Comfill** and then click the form.

Put the name of the database field or variable that you want to associate with this control into the **Input** property.

Best practice

Use image id=7 for fill boxes. Find and Fill buttons should not use the down arrow icon, because it makes the Find or Fill look like a Combo Box.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Unused.
Caption Condition	Unused.
Input	Specify the database field or variable to associate with this control.
Accessible Name	Specify a component name for use with accessibility software. (Optional) **Note** Note This function is only supported by the Web client. If no name is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, it uses the component's text. Accessibility software can query the Web client to get this information and use it to present data to the user in a variety of ways. For example, speech simulation software says the name and the type of component when the component has focus.

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note** Note This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.
Read-Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to true.
Mandatory	Select this option so that a red asterisk indicating a required field is displayed. This is a visible change only. - To make the field mandatory for any form it appears on, use the System Definition Utility. - To make the field mandatory for a small number of forms only, use Format Control.
Mandatory Condition	Specify an expression to override the Mandatory property when that expression evaluates to true.

Mandatory For Read-Only	If you set this option to true, the system displays the mandatory icon on the read-only Comfill widgets. You must add appropriate accessible descriptions to guide users how to fill the value for the mandatory and read-only fields. **Note** Note This function is only supported by the Web client.
Array Length	Specify the size of the scrolling region used to view array entries. A scroll bar appears beside the fields to allow users to view the array entries. • If a field is assigned an Array Length of 5, the form stacks five fields vertically to allow users to view the five array entries. • If the Input data type is scalar, only a single text box appears. **Note** Note The screen object must be associated with an array data structure. The default is 0, which means one vertical line of information appears. You can set this value to any positive integers.
Password	Select this option to make the text entered in the field a series of asterisks on the screen. The data is not encrypted, just transposed.
Maximum Chars	Specify the number of characters the user can enter in the field. The default is 0 (unlimited).
Maximum Characters Beep	Select this option to notify the user by sounding a beep that they reached the maximum number of characters.
Case Conversion	Specify how to change the case of text entered in the field. The choices are None, Upper, Lower.
Decimals	Specify the number of decimal points for numbers in this field. Service Manager modifies numbers entered in the field to the number of decimal points you specify.
Parse	Select this option to verify that the text entered in a field is the correct syntax. The syntax for the field is defined by the type of field it displays (date, expression, number, and so forth).

Data Changed Event	Specify the option number (such as Button ID) to call if the data contained in the object changed. This sends an event to the display RAD application.
Value List	In conjunction with the Input property, defines how the Display List values are identified in the database. Value and Display Lists are entered using the Edit List dialog box. You can enter hard coded entries for each list, or you can supply a variable as the first and only entry. The run time values of the variable are used to populate these lists. This property is supported only in association with the toolbar configuration Knowledge Management Extended, which is not intended to be used outside of the out-of-box Knowledge Management module. **Note**: When defining Value List and Display List properties, you should avoid using the following reserved characters: • semicolon (;) • tab key • newline (carriage return) • form feed • backspace • equal sign (=) **Important**: Do not use keys (like backspace), as the system can't process their ASCII representation entries correctly which causes unpredictable results.
Value List Condition	Specify an expression to override the property when that expression evaluates to true.

Display List	Specify the values that appear in the drop-down list at run time. There must be a one–to–one correspondence between the values for Value List and for Display List. If the Display List is the only populated property, the display values are written to the database. This property is supported only in association with the toolbar configuration Knowledge Management Extended, which is not intended to be used outside of the out-of-box Knowledge Management module. **Note**: When defining Value List and Display List properties, you should avoid using the following reserved characters: • semicolon (;) • tab key • newline (carriage return) • form feed • backspace • equal sign (=) **Important**: Do not use keys (like backspace), as the system can't process their ASCII representation entries correctly which causes unpredictable results.
Display List Condition	Specify an expression to override the Display List property, when the expression evaluates to True.
Box Lines	Specify the window size of the drop-down list.
Select Only	Selecting the checkbox requires the user to select from the Combo Box drop-down list. If the user makes a manual entry, the value must already exist in the drop-down list (defined in the Value List and Display List properties). Before you select this option, make sure that the Combo Button Visible option is also selected. Clearing the checkbox allows the user to select from the drop-down list or to manually enter a value.
Input Conversion	Specify the name of an Input Conversion routine. This property is used by RAD subroutines that mask the display of data on an input field, or check and validate the entry of data into an input field. If specified, the routine is called at runtime to convert the contents of the field before it is displayed on the screen. For more information, refer to RAD input and output conversion.

Mask	Specify the mask value (the condition that sets a value to be passed for this field to the input and output conversion routines). When a field has a mask value and you save the record, the format of the mask value displays in the field. For example, you decide to update the purchase cost information of a contract in Configuration Management. You enter a date in the Currency EX Date field of the Financial tab as 02/26/09. The condition for the mask value is set as: **xxx dd,yyyy**. When you save the record, the Currency Ex Date field displays the date format of February 26, 2009. Valid values for the mask vary depending on the routine being called.
Output Conversion	Specify the name of an Output Conversion routine. This property is used by RAD subroutines that mask the display of data on an output field, or check and validate the entry of data into an input field. If specified, the routine is called at runtime to convert the contents of the field before it is displayed on the screen. For more information, refer to RAD input and output conversion.
Combo Button Visible	Select this option to make the Combo Button visible when the form opens.
Combo Button Visible Condition	Specify an expression to override the ComboButton Visible property when that expression evaluates to true.
Fill Button ID	Specify a Control ID to transmit when clicked. For example, 3 corresponds to **F3**, and 0 corresponds to **Enter**.
Fill Button Image	Specify an icon to display on the field, if desired. You can select from 1 to 17. The default value is 7.
Fill Button Visible	Select this option to make the Fill Button visible when the form opens.
Fill Button Visible Condition	Specify an expression to override the Fill Button Visible property when that expression evaluates to true.
Third Button ID	Specify a Control ID to transmit when clicked. For example, 3 corresponds to **F3**, and 0 corresponds to **Enter**.
Third Button Image	Specify an icon to display on the field, if desired. You can select from 1 to 17. The default value is 10.
Third Button Visible	Select this option to make the Third Button visible when the form opens.
Third Button Visible Condition	Specify an expression to override the Third Button Visible property when that expression evaluates to true.
Popup Subform Format	Specify the form to display.
Popup Subform Input	Specify the database field or variable to associate with this control.

Popup Subform Enabled	Select this option to enable the Popup Subform for this control.
Popup Subform Enabled Condition	Specify an expression to override the Popup Subform Enabled property when that expression evaluates to true.
Auto Complete	Enable the auto complete feature for this control. **Note** Note This setting takes effect only when the comfillAutoComplete parameter is set to **true**.

Related topics

Creating and editing forms
Using the drawing canvas
Forms Designer properties view
Using pop-ups
Forms Designer best practices
Access Forms Designer
Create a form using the Form Wizard
Update a form
Add a subform to a form
Checkbox control
Combo Box control
Radio button control
Subform control
Forms Designer controls and tools

1.1.1.2.10. Convert Form Layout tool

This button converts all of the fields on the left side of the form into one group, and converts each of the notebook tabs on the right into groups, arranged vertically, under the left side fields. These groups are collapsible when displayed in the Web client.

When you click this button, a dialog box will display to inform you that the action will modify the form layout to use collapsible sections and that this can't be undone after you have saved the form. If you click **Yes** to proceed, the new layout is displayed. All of the fields and controls that were on the left side of the form are now contained in one Group. Clicking at the top of the group displays the Properties for the group. You will see that the following checkboxes are already selected:

- Floating group enabled: The group to change its relative position on the form so that it's not hidden when sections above it expand downward, and doesn't leave a blank spot after being collapsed.
- Collapse enabled: The group is displayed as a collapsible group that can be expanded and collapsed by the user.
- Default to Expanded: The collapsible group is expanded (open), by default. The user can close it if desired.

You can scroll down to see that the fields that were previously contained in the first (leftmost) notebook tab are also now contained in a Group. The Default to Expanded checkbox is not selected, so that this group will display as collapsed by default, until the user opens it.

If you scroll down further you will see the contents of each of the notebook tabs, previously displayed from left to right, are now displayed as groups, from top to bottom.

If you want to change the order of the sections, you can do so by changing the Y coordinate, which controls the vertical placement of the group on the form.

Related topics

Form creation
Forms Designer
Creating and editing forms
Using the drawing canvas
Forms Designer properties view
Setting properties
Forms Designer best practices
Building accessible forms
Access Forms Designer
Create a form using the Form Wizard
Update a form
Forms Designer controls and tools
Grid view tool
Selection tool

1.1.1.2.11. Date control

Use this control to add a text box that displays and enables users to update the contents of a date field.

To place a date field on a form, click **Date** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Accessible Name	Specify a component name for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no name is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, it uses the component's text. Accessibility software can query the Web client to get this information and use it to present data to the user in a variety of ways. For example, speech simulation software says the name and the type of component when the component has focus.

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Input Conversion	Specify the name of an Input Conversion routine. This property is used by RAD subroutines that mask the display of data on an input field, or check and validate the entry of data into an input field. If specified, the routine is called at runtime to convert the contents of the field before it is displayed on the screen. For more information, refer to RAD input and output conversion.
Mask	Specify the mask value (the condition that sets a value to be passed for this field to the input and output conversion routines). When a field has a mask value and you save the record, the format of the mask value displays in the field. For example, you decide to update the purchase cost information of a contract in Configuration Management. You enter a date in the Currency EX Date field of the Financial tab as 02/26/09. The condition for the mask value is set as: **xxx dd,yyyy**. When you save the record, the Currency Ex Date field displays the date format of February 26, 2009. Valid values for the mask vary depending on the routine being called.
Output Conversion	Specify the name of an Output Conversion routine. This property is used by RAD subroutines that mask the display of data on an output field, or check and validate the entry of data into an input field. If specified, the routine is called at runtime to convert the contents of the field before it is displayed on the screen. For more information, refer to RAD input and output conversion.

Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.
Read-Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to true.
Parse	Select this option to verify that the text entered in a field is the correct syntax. The syntax for the field is defined by the type of field it displays (date, expression, number, and so forth).
Mandatory	Select this option so that a red asterisk indicating a required field is displayed. This is a visible change only. - To make the field mandatory for any form it appears on, use the System Definition Utility. - To make the field mandatory for a small number of forms only, use Format Control.
Mandatory Condition	Specify an expression to override the Mandatory property when that expression evaluates to true.
Array Length	Specify the size of the scrolling region used to view array entries. A scroll bar appears beside the fields to allow users to view the array entries. - If a field is assigned an Array Length of 5, the form stacks five fields vertically to allow users to view the five array entries. - If Array Length is set to -1, the form dynamically creates as many fields as there are entries in the array. - If the Input data type is scalar, only a single text box appears. **Note**: The screen object must be associated with an array data structure. The default is 0, which means one vertical line of information appears.

Type	Specify the format of the date value. There are three formats available: the date/time format (such as 2023/1/1, 01:01:01, or 2023/1/1 01:01:01) date format (such as 2023/1/1), and the duration time format (such as 1 01:01:01). When this Type value is empty, both the date/time format and the duration time format are allowed. When this Type value is set as **Datetime**, only the date/time format is allowed. When this Type value is set as **Date-only**, only the date format is allowed. When this Type value is set as **Duration**, only the duration time format is allowed.
Time Zone	Specify the time zone for the date set in the Date control widgets. For example, suppose there is a Planned Start Date control on a form and you have set the Time Zone property of this control to `US/Mountain`. The Planned Start Date field displays the US/Mountain time zone as you specified instead of the operator's actual time zone. **Note:** This Property works no matter the **Duration** property is selected or not.
Time Zone Field	Specify the field that indicates the time zone for the date set in the duration property.
Data Changed Event	Specify the option number (such as Button ID) to call if the data contained in the object changed. This sends an event to the display RAD application.

1.1.1.2.12. Decimal control

Use this control to add an input field that accepts only numeric input and offers spinner buttons or a slider to increase or decrease a value.

To place a decimal field on a form, click **Decimal** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Accessible Name	Specify a component name for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no name is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, it uses the component's text. Accessibility software can query the Web client to get this information and use it to present data to the user in a variety of ways. For example, speech simulation software says the name and the type of component when the component has focus.

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Input Conversion	Specify the name of an Input Conversion routine. This property is used by RAD subroutines that mask the display of data on an input field, or check and validate the entry of data into an input field. If specified, the routine is called at runtime to convert the contents of the field before it is displayed on the screen. For more information, refer to RAD input and output conversion.
Mask	Specify the mask value (the condition that sets a value to be passed for this field to the input and output conversion routines). When a field has a mask value and you save the record, the format of the mask value displays in the field. For example, you decide to update the purchase cost information of a contract in Configuration Management. You enter a date in the Currency EX Date field of the Financial tab as 02/26/09. The condition for the mask value is set as: **xxx dd,yyyy**. When you save the record, the Currency Ex Date field displays the date format of February 26, 2009. Valid values for the mask vary depending on the routine being called.
Output Conversion	Specify the name of an Output Conversion routine. This property is used by RAD subroutines that mask the display of data on an output field, or check and validate the entry of data into an input field. If specified, the routine is called at runtime to convert the contents of the field before it is displayed on the screen. For more information, refer to RAD input and output conversion.

Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.
Read-Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to true.
Decimals	Specify the number of decimal points for numbers in this field. Service Manager modifies numbers entered in the field to the number of decimal points you specify.
Parse	Select this option to verify that the text entered in a field is the correct syntax. The syntax for the field is defined by the type of field it displays (date, expression, number, and so forth).
Mandatory	Select this option so that a red asterisk indicating a required field is displayed. This is a visible change only. - To make the field mandatory for any form it appears on, use the System Definition Utility. - To make the field mandatory for a small number of forms only, use Format Control.
Mandatory Condition	Specify an expression to override the Mandatory property when that expression evaluates to true.
Maximum Value	Set the maximum value that a spinner button or a slider can reach.
Minimum Value	Set the minimum value that a spinner button or a slider can reach.
Step	Specify a large step value to be used when pressing Page Up or Page Down. The default value is 5. **Note**: In this release, this value applies to the slider view only.
Slider View	Select this option to display the decimal control as a slider. If this option is not selected, the system displays the decimal control as spinner buttons. **Note**: You must sepcify the **Maximum Value** and the **Minimum Value** for the slider view. If these two fields are blank, the system uses the default values of 100 and 0, respectively. **Note**: The web client accessible mode doesn't support the slider view. In addition, the right-to-left languages doesn't support the slider view as well.

1.1.1.2.13. Dynamic Form control

Use this control to add a dynamic form that becomes visible when populated by XML from a RAD application or JavaScript. The aspect and content of the dynamic form depend on the XML.

To place a dynamic form on a form, click **Dynamic Form** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

To view the dynamic form, open the form on which it is located with Database Manager. Each time you save changes in Forms Designer, exit the form and reload it in Database Manager.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.
Read-Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Content Always Visible	If this option is selected, all the content inside the dynamic form will be displayed, despite the visible condition set for each widget. Usually this property is set to true only for a dynamic form in a preview page.

1.1.1.2.14. Embedded Viewer control

Use this control to add an Embedded Viewer that enables Web client users to view integrations.

To place an Embedded Viewer on a form, click **Embedded Viewer** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note:** This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Frameborder	Specify whether a border is displayed around the embedded viewer. If you don't specify anything or enter 1, a border is displayed. If you enter 0, there is no border.
Height	Specify the height of the object in alignment grid units.
Input	Specify the database field or variable to associate with this control.
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Use client styling	Select this option to display content with the font, font size, margins and background color set in the client.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Width	Specify the width of the object in alignment grid units.
X	Specify the object's horizontal position based on the left edge of the object.

Y	Specify the object's vertical position based on the top edge of the object.

1.1.1.2.15. File control

Use this control to enable users to launch a File Open dialog box and upload or download files.

To place a file on a form, click **File** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Accessible Name	Specify a component name for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no name is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, it uses the component's text. Accessibility software can query the Web client to get this information and use it to present data to the user in a variety of ways. For example, speech simulation software says the name and the type of component when the component has focus.

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Mandatory	Select this option so that a red asterisk indicating a required field is displayed. This is a visible change only. - To make the field mandatory for any form it appears on, use the System Definition Utility. - To make the field mandatory for a small number of forms only, use Format Control.
Mandatory Condition	Specify an expression to override the Mandatory property when that expression evaluates to true.
Transfer direction	Specify whether you want the user to be able to download or upload the file. The default is download. **Note**: If you don't have the transfer direction set properly, the user will get an error message and will not be able to upload/download the file. For example, if pressing the Specify File button gets the following error message, "File not found. Please verify the correct file name was given.", you may need to change the transfer direction from upload to download.
File Directory	Specify the absolute path to a directory. For example, c:/temp. If no directory is specified, the Service Manager RUN directory defaults.
File Type	Specify the MIME–type of the file the File Open dialog prompts for. For example, `.txt` for text files. If no MIME type is specified, the default is `All Files (*.*)`.
Title	Specify the name of the file you want to transfer.

1.1.1.2.16. Frame control

Use this control to add a container that creates a cosmetic rectangular border. Use bevel frames to make your forms more attractive or easier to understand.

To place a frame on a form, click **Frame** and then click the form.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Background Color	Specify the Background Color. You must select the Opaque property checkbox for the Back Color to have an effect. The only Background Color available for a Windows client marquee is black.
Outer Bevel Width	Specify the amount, in points, to increase or decrease the standard font size.
Inner Bevel Width	Specify an expression to override the Font Increase property when that expression evaluates to true.

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1.1.1.2.17. Graph control

Use this control to display a diagram of connected nodes representing a system of two or more interrelated things. To add a graph to a form, either click on the control in the toolbar and then click on the drawing canvas, or, drag and drop the control from the toolbar to the drawing canvas.

Note

Note

Service Manager discontinues the support of Internet Explorer 11. The previous Applet-sensitive features (including telephony (CTI), workflows, and CI visualization) in Service Manager have alternative solutions or new implementations.

Therefore, the implementation of graph control has changed as well. The embedded toolbar in the graph control is slightly different between the Windows client and the web client. In the Windows client, the graph control provides a toolbar that is more comprehensive, including Zoom In, Zoom Out, Fit Content, Select Mode, Windows Area Mode, View Drag Mode, Drill Down Mode, Overview, Show Node Labels, Show Edge Labels, Print Preview, and a layout selection drop-down list. In the web client, the graph control provides a toolbar that includes Zoom In, Zoom Out, Fit Content, Show Overview, and Show Edge Labels only. The web client displays all base CI images hierarchically.

Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Input	Specify the database field or variable to associate with this control. The primary use for this property is to set the focus on the graph.
Accessible Description	(Optional) Specify a component description for use with accessibility software. **Note:** This function is only supported by the web client. If no description is present, the web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the web client will use the component's text. Accessibility software can query the web client to get this information and use it to present it to the user in a variety of ways. For example, when a component gets the focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.

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1.1.1.2.18. Grid view tool

This button enables you to view the alignment grid while designing a form.

Click the **Grid Preview** icon to overlay the design area with a grid.
Click the icon again to remove the overlay.

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1.1.1.2.19. Group control

Use this control to add a container that enables you to logically group associated items. In the Windows client the Group has a rectangular border with a text label at the top.



Note

Note In the Web client, a group can be displayed as a collapsible section; see the last three properties listed in the table below.



Caution

Caution When designing forms with collapsible sections, use the following guidelines to avoid formatting problems when the form is displayed in the Web client.

- All groups on the form should be marked as Collapsible-enabled.
- All widgets must be placed in a collapsible group box; no widgets should remain floating independently on the form.
- Do not stack or mix collapsible group boxes with non-collapsible group boxes.

To place a group on a form, click **Group** and then click the form.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Foreground Color	Select the text color from the drop-down list.
Foreground Color Condition	Specify a formula to evaluate the ForeColor property at run time. The result of this evaluation overrides the value assigned to the property.
Background Color	Specify the Background Color. You must select the Opaque property checkbox for the Back Color to have an effect. The only Background Color available for a Windows client marquee is black.
Background Color Condition	Specify an expression to override the BackColor property when that expression evaluates to true.
Font	Select a font for the text of your label, table, timer, or marquee caption from the drop-down list.
Bold	Select this option to apply bold to the label, default text, frame caption, or marquee caption.
Bold Condition	Specify an expression to override the Bold property when that expression evaluates to true.
Italic	Select this option to apply italics to your label, default font, or caption.

Italic Condition	Specify an expression to override the Italic property when that expression evaluates to true.
Heading Level	Specify the heading level for this control so that a corresponding HTML heading tag (<h1> to <h6>) is generated. **Note** Note This function is only supported by the Web client.
Justification	Specify the justification for the object label as Left, Right, or Center.
Font Increase	Specify the amount, in points, to increase or decrease the standard font size.
Font Increase Condition	Specify an expression to override the Font Increase property when that expression evaluates to true.
Array Length	Specify the size of the scrolling region used to view array entries. A scroll bar appears beside the fields to allow users to view the array entries. • If a field is assigned an Array Length of 5, the form stacks five fields vertically to allow users to view the five array entries. • If Array Length is set to -1, the form dynamically creates as many fields as there are entries in the array. • If the Input data type is scalar, only a single text box appears. **Note** **Note**: The screen object must be associated with an array data structure. The default is 0, which means one vertical line of information appears.
Floating group enabled	This option enables the group to change its relative position on the form so that it's not hidden when sections above it expand downward, and doesn't leave a blank spot after being collapsed. **Note** Note This function is only supported by the Web client.

Collapse enabled	This option enables the group to be displayed as a collapsible section that can be expanded and collapsed by the user. **Note** Note This function is only supported by the Web client.
Default to Expanded	Displays a collapsible group as expanded, by default. The user can close it if desired. **Note** Note This function is only supported by the Web client.
Radio Group	If you set this option to true, the system regards the Group widget as a radio group and you are only allowed to add radio buttons to the Group widget. In addition, you must set identical values for the Input property of these radio buttons. If set this option to false, ignore the following three properties as described in this table. **Note** Note This function is only supported by the Web client.
Radio Group Orientation	Specify the aria-orientation attribute for the Radio Group. If not specified, the orientation is assumed to be vertical. **Note** Note This function is only supported by the Web client.
Mandatory	Setting Mandatory to **Yes** by checking the box indicates that the Radio Group is required and therefore a red asterisk indicating a required field is displayed. This is a visible change only. **Note** Note This function is only supported by the Web client.

Mandatory Condition	Specify an expression to override the Mandatory property when that expression evaluates to true. **Note** Note This function is only supported by the Web client.

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1.1.1.2.20. HTML Editor control

Use this control to add an HTML Editor that enables users to add standard HTML to the form. Service Manager displays the rendered HTML in the HTML Viewer.

To place an HTML Editor on a form, click **HTML Editor** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Important: For Accessibility, the HTML Editor must be the last widget in its form or page, because you will not be able to tab to any widgets placed under it.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.

Toolbar Configuration	Enables you to select an HTML Editor Toolbar from the available toolbars. - **Standard** includes options for the most common HTML tags that enable you to add font style (bold, italic, color, and underscore) and hyperlinks to texts. - **Knowledge Management** includes options for the HTML tags required for Knowledge Management plus the Standard tags. - **Knowledge Management Extended** includes options Knowledge Management development, plus the Knowledge Management and Standard tags.
Insert Image Option	This property is for internal Service Manager development use only. It functions properly only in association with appropriate application code.
Insert Attachment Option	This property is for internal Service Manager development use only. It functions properly only in association with appropriate application code.
Insert Link Option	This property is for internal Service Manager development use only. It functions properly only in association with appropriate application code.
Value List	In conjunction with the Input property, defines how the Display List values are identified in the database. Value and Display Lists are entered using the Edit List dialog box. You can enter hard coded entries for each list, or you can supply a variable as the first and only entry. The run time values of the variable are used to populate these lists. This property is supported only in association with the toolbar configuration Knowledge Management Extended, which is not intended to be used outside of the out-of-box Knowledge Management module.
Display List	Specify the values that appear in the drop-down list at run time. There must be a one–to–one correspondence between the values for Value List and for Display List. If the Display List is the only populated property, the display values are written to the database. This property is supported only in association with the toolbar configuration Knowledge Management Extended, which is not intended to be used outside of the out-of-box Knowledge Management module.

1.1.1.2.21. HTML Viewer control

Use this control to add an HTML Viewer that enables users to view the HTML created using the HTML Editor.

To place an HTML Viewer on a form, click **HTML Viewer** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input**

property.

Warning: Keep in mind that text you copy and paste from other sources may contain formatting tags, even if it looks like plain text.

Property	Usage
Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note:** This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Height	Specify the height of the object in alignment grid units.
HTML-Generating Script	A Javascript function that generates the content for the HTML Viewer. For example, open the form "configurationItem" in Forms Designer. On the "CI Changes" notebook page there is an HTML Viewer which has "configurationManagement.getPendingChanges" for that property. To see this function, type **sl** in the command line, and then type **configurationManagement** in the Name field. Click **Search**, and then scroll down. You will see the "getPendingChanges" function.
Input	Specify the database field or variable to associate with this control.
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.

Use client styling	Select this option to display an HTML snippet. The viewer will display the text respecting the font, font size, margins and background color set in the client. A fully defined HTML document may not display correctly. **Note:** Mark the checkbox for Service Catalog. Clear the checkbox to display a fully defined HTML document. A fully defined HTML document contains &ltl> <head></head><body></body></html>. If the input doesn't contain any HTML formatting it will display plain text. **Note:** Clear the checkbox for Knowledge Management.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Width	Specify the width of the object in alignment grid units.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Frameborder	Specify whether a border is displayed around the HTML viewer. If you don't specify anything or enter 1, a border is displayed. If you enter 0, there is no border.
Standards Mode	Select this option if you want the web browser to render the HTML document in Standards Mode. When this option is selected, a DOCTYPE is added to the beginning of the document.

1.1.1.2.22. Image control

Use this control to add an image from a Microsoft Windows image file.

To place an image on a form, click **Image** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.
Read-Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to true.

Image	Specifies the name of an image file to display on the control. The image file must reside in any of the following paths, in which Service Manager looks for the image: **Windows client**: • <Service Manager>\Client\plugins\com.hp.ov.sm.client.common_x.xx.xxx\src\resources\icons • A local directory on the client machine, which is specified in the **Image path** field (**Window** > **Preferences**). For example: C:\smCustomImages. **Web client**: • <Service Manager web tier>\images. For example: <Tomcat>\webapps\webtier-x.xx\images. **Note** Note When you specify the image file for this property, you need to provide both the file name and the file name extension.
Image Condition	Specify an expression to override the Image property when that expression evaluates to true.

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1.1.1.2.23. Journal Updates Viewer control

Use this control to add a Journal Updates Viewer that enables users to view the journal updates.

To place a Journal Updates Viewer on a form, click **Journal Updates Viewer** and then click the form.

Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button, or checkbox, you will almost always use the default of 0.
Accessible Description	(Optional) Specify a component description for use with accessibility software. **Note:** This function is only supported by the web client. If no description is present, the web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the web client will use the component's text. Accessibility software can query the web client to get this information and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Login Name Conversion	Select this option to display the operator's Full Name for updates. If not selected, the operator's name will be displayed according to the environment settings.

1.1.1.2.24. Label control

Use this control to add a label. A label is a single line of text you can use to give titles to forms, give labels to objects within the form, or otherwise place text on the form.

To place a label on a form, click **Label** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input**

property.

Best practices

- Use black for label text.
- Left-align label text and end it with a colon.
- Do not create colored text "drop shadows." These don't render well in the Web client.
- Labels on menu forms (for example, menu.gui.home) should have a font increase of 2.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Label for	Specify the name of a control that uses this label. If you want to use this label for multiple controls, you need to specify semicolon-separated control names. **Note** Note If you specify a variable for this label, you need to provide a valid value here. Otherwise, the label can't be read by JAWS.
Foreground Color	Select the text color from the drop-down list.
Foreground Color Condition	Specify a formula to evaluate the ForeColor property at run time. The result of this evaluation overrides the value assigned to the property.
Background Color	Specify the Background Color. You must select the Opaque property checkbox for the Back Color to have an effect. The only Background Color available for a Windows client marquee is black.
Background Color Condition	Specify an expression to override the BackColor property when that expression evaluates to true.

Background Style	Applies to the display of the label in the Web client only. Select **Underline** to display the label as underlined text. Select **Solid Background** to display the label background as a solid colored bar.
Font	Select a font for the text of your label, table, timer, or marquee caption from the drop-down list.
Bold	Select this option to apply bold to the label, default text, frame caption, or marquee caption.
Bold Condition	Specify an expression to override the Bold property when that expression evaluates to true.
Italic	Select this option to apply italics to your label, default font, or caption.
Italic Condition	Specify an expression to override the Italic property when that expression evaluates to true.
Font Increase	Specify the amount, in points, to increase or decrease the standard font size.
Font Increase Condition	Specify an expression to override the Font Increase property when that expression evaluates to true.
Array Length	Specify the size of the scrolling region used to view array entries. A scroll bar appears beside the fields to allow users to view the array entries. • If a field is assigned an Array Length of 5, the form stacks five fields vertically to allow users to view the five array entries. • If Array Length is set to -1, the form dynamically creates as many fields as there are entries in the array. • If the Input data type is scalar, only a single text box appears. The default value is 0, which means one vertical line of information appears. **Note** Note The screen object must be associated with an array data structure.

Heading Level	Specify the heading level for this control so that a corresponding HTML heading tag (<h1> to <h6>) is generated. **Note** Note This function is only supported by the Web client.
Justification	Specify the justification for the object label as Left, Right, or Center.

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1.1.1.2.25. Link label control

Use this control to add a label that includes a caption and an icon that links to another source.

To place a link label on a form, click **Link label** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Accessible Name	Specify a component name for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no name is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, it uses the component's text. Accessibility software can query the Web client to get this information and use it to present data to the user in a variety of ways. For example, speech simulation software says the name and the type of component when the component has focus.

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Button ID	Specify a numeric identification that specifies a Control ID to transmit when clicked.
Button ID Condition	Specify an expression to override the Button ID property when that expression evaluates to true.
Image	Specifies the name of an image file to display on the control. The image file must reside in any of the following paths, in which Service Manager looks for the image: **Windows client**: • <Service Manager>\Client\plugins\com.hp.ov.sm.client.common_x.xx.xxx\src\resources\icons • A local directory on the client machine, which is specified in the **Image path** field (**Window** > **Preferences**). For example, C:\smCustomImages. **Web client**: • <Service Manager web tier>\images\obj16 **Note** Note When you specify the image file for this property, you need to provide both the file name and the file name extension.
Description	Specify the description text for the label.

Balloon Help	Specify the text string to display when the cursor is held over the button to offer more information about the button's function.
Button Action	Specify an action for the button. The available options is **Open Smart Search Widget**.
Button Action Params	Specify a parameter for the button action. **Note** Note The button action that's currently available doesn't need a parameter.
Enabled	Specify an initial state.
Enabled Condition	Specify a condition that enables the button when a table row is selected. Example: [%tablename] == "selected" or [%tablename] != "selected" where tablename is the value of the Name property of the table on the screen.

1.1.1.2.26. List Builder control

Use this control to add a compound widget that allows the user to construct a list of choices.

To place a List Builder on a form, click **List Builder** and then click the form.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Accessible Name	Specify a component name for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no name is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, it uses the component's text. Accessibility software can query the Web client to get this information and use it to present data to the user in a variety of ways. For example, speech simulation software says the name and the type of component when the component has focus.

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Allow Duplicates	Click the box to allow duplicate items.
Choices Name	Specify a label for the Choices list.
Display List	Specify the values that appear in the drop-down list at run time. There must be a one–to–one correspondence between the values for and for Display List. If the Display List is the only populated property, the display values are what is written to the database.
Display List Condition	Specify an expression to override the Display List property when that expression evaluates to true.
Selection Name	Specify a label for the Selections list.
Sortable	Click the box to allow reordering of the list items.
Value List	In conjunction with the Input property, defines how the Display List values are identified in the database. Value and Display Lists are entered using the Edit List dialog box. You can enter hard coded entries for each list, or you can supply a variable as the first and only entry. The run time values of the variable are used to populate these lists. This property is supported only in association with the toolbar configuration Knowledge Management Extended, which is not intended to be used outside of the out-of-box Knowledge Management module.
Value List Condition	Specify an expression to override the Value List property when that expression evaluates to true.

1.1.1.2.27. Marquee control

Use this control to add an attention–getting text message that scrolls continuously across the screen from right to left.

To place a marquee on a form, click **Marquee** and then click the form. Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Mozilla-based browsers don't support the marqee tag and thus can't display them in Web client forms.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Foreground Color	Select the text color from the drop-down list.
Background Color	Specify the Background Color. You must select the Opaque property checkbox for the Back Color to have an effect. The only Background Color available for a Windows client marquee is black.
Font	Select a font for the text of your label, table, timer, or marquee caption from the drop-down list.
Bold	Select this option to apply bold to the label, default text, frame caption, or marquee caption.
Italic	Select this option to apply italics to your label, default font, or caption.
Font Increase	Specify the amount, in points, to increase or decrease the standard font size.

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1.1.1.2.28. Notebook tab control

Use this control to add a tab to an existing notebook. Navigate to each page of the notebook by selecting its tab.

Click to the right of the last tab on the notebook to open the notebook's properties. Click the notebook tab and then the blank area below the tab to open the notebook tab's properties.

Best practices

Forms with tabs should include a header section which is not part of the tab. This header should include important description information about the record. For example, incident records should show, at a minimum, the incident number.

To place a tab on a notebook, click **Notebook tab** and then click the notebook.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.

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1.1.1.2.29. Notebook control

Use this control to subdivide the contents of a screen into logical groups or categories. Notebooks provide an aesthetic way of organizing large amounts of data into small spaces.

Click to the right of the last tab on the notebook to open the notebook properties. Click the tab and then the blank area below the tab to open the notebook tab properties.

To place a notebook on a form, click **Notebook** and then click the form.

Best practice

Do not place notebooks next to other form elements or other notebooks.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Group rendering enabled	Causes the notebook to be displayed as a group rather than a tabbed notebook. The contents of each tab are displayed as separate sections of the group. The notebook tabs that were arranged from left to right inside the notebook are displayed from top to bottom in the group.
Lock rendering style	Adds the ability to switch between the Group style and the Tab style for a Notebook control in the web client.
Heading Level	Specify the heading level for this control so that a corresponding HTML heading tag (<h1> to <h6>) is generated. **Note** Note This function is only supported by the Web client.

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1.1.1.2.30. Radio Button control

Use this control to add a radio button to a form. A radio button enables the user to select one value from a set of mutually exclusive values. The items in the set are associated with a database field or variable. Use radio buttons to give the user a quick way to choose one option from a set of fixed and mutually exclusive string values. It is a good practice to group related radio buttons within a frame.

Note: Radio buttons aren't supported on record lists.

Best practices[edit]

- If the choice is logical (true or false), use a checkbox instead.
- If choices are extensive or if the user may need to enter a choice manually, use a Combo Box instead.

To place a radio button on a form, click **Radio button** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Accessible Name	Specify a component name for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no name is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, it uses the component's text. Accessibility software can query the Web client to get this information and use it to present data to the user in a variety of ways. For example, speech simulation software says the name and the type of component when the component has focus.

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Foreground Color	Select the text color from the drop-down list.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.
Read-Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to true.
Mandatory	Select this option so that a red asterisk indicating a required field is displayed. This is a visible change only. - To make the field mandatory for any form it appears on, use the System Definition Utility. - To make the field mandatory for a small number of forms only, use Format Control.
Mandatory Condition	Specify an expression to override the Mandatory property when that expression evaluates to true.
Data Changed Event	Specify the option number (such as Button ID) to call if the data contained in the object changed. This sends an event to the display RAD application.
Value	Specify the value to assign to the field or variable specified in the Input property when a user selects this radio button.

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1.1.1.2.31. Selection tool

This button enables you to select, deselect, and modify design objects.

To select a control, click **Selection tool** and then click the control.

Switch from selection mode to creation mode

If you want to create multiple instances of an object, hold down the **Ctrl** key and click anywhere on the drawing canvas.

Select one object

Click on the object. Grab points are displayed indicating that the object is selected.

Or, hold down the left mouse button and drag a lasso across any part of the object. After lassoing the object, handles are displayed.

Select multiple objects

Hold down the **Ctrl** key and click on multiple objects one at a time. Grab points are displayed on each object. The properties view displays the common properties of the objects selected.

Or, drag and release a lasso across any part of the desired objects. After lassoing the objects, handles are displayed on each object.

Clear one object

Hold down the **Ctrl** key and click on the object. Grab points disappear, indicating that the object is no longer selected.

Clear multiple objects

Click on a blank area of the drawing canvas.

Position one object

Click on the object and continue holding down the left mouse button. Drag the object to the desired position.

Position multiple objects

Hold down the **Ctrl** key and one at a time, click on the multiple objects you want to position. The movement cursor appears when you hover over the selected objects.

At this point, you can drag and drop the objects as a group anywhere on the drawing canvas.

Or, hold down the **Ctrl** key and one at a time, click on the multiple objects you want to position. Then enter the X or Y coordinate for all the items in the properties editor. For example, if you want to position multiple objects with the same left border, set the X property. The Y property can be used in a similar way to position objects with the same top border.

Resize an object

Select the object. Drag the object handles to resize the object.

Delete one or more objects

Select the object(s). Press the **Delete** key.

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1.1.1.2.32. Script control

Use this control to add a color-coded syntax editor to a form. This enables the user to enter JavaScript.

To place a script on a form, click **Script** and then click the form.

Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Accessible Name	Specify a component name for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no name is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, it uses the component's text. Accessibility software can query the Web client to get this information and use it to present data to the user in a variety of ways. For example, speech simulation software says the name and the type of component when the component has focus.

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note** Note This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Script	Specify the type of script. The default is *js* for JavaScript. The Script Control on the Windows client has the following features: • Content assist • Syntax highlighting • Quick outline • Hyperlink • Keyboard shortcuts • Folding The Script Control on the web client has the following features: • Syntax highlighting • Keyboard shortcuts

1.1.1.2.33. Subform control

Use this control to display a subform on another form.

Subforms displayed on forms have full functionality. Subforms displayed as pop-ups are read-only; users can't interact with them. To display the subform as a pop-up, use the Comfill control instead.

To place a subform on a form, click **Subform** and then click the form.

Put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Format	Specify the form to display. Be sure you type the exact form name.
Virtual Join	Select this option to associate virtual join run time processing with the subform object. Clear the checkbox to use a same-file join.
Display Blank	Select this option to display the subform even if no records were found for the virtual join. Clear the checkbox to skip display of the format if there are no matching records. **Note**: Used only for virtual joins.
Display Using Table	Select this option to display the subform in table format.
Input	Specify the database field or variable to associate with this control.

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1.1.1.2.34. Table column control

Use this control to add a column to an existing table.

> **Note**
>
> Note To avoid unpredictable behavior, don't set a DVD condition to use a variable that's based on a record displayed in the Table widget.

To add a column to a table, click **Table column** and then click the table.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Best practice

Column headings should follow book title capitalization rules, for example, File Name not file name or file.name.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Accessible Name	Specify a component name for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no name is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, it uses the component's text. Accessibility software can query the Web client to get this information and use it to present data to the user in a variety of ways. For example, speech simulation software says the name and the type of component when the component has focus.

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Field	In the case where the array specified in the input property is an array of structure, it specifies the field in the structure for this object.
Column Width Percent	Enables you to set the column width to a percentage of the page.
Minimum Column Width	Enables you to set the minimum width of the column in a table.
Clickable Title	Select this option to allow the user to click the title.
Button ID	When an individual column Button ID is set to 0, the Default Button ID is used.
Show Title	Select this option to make the title appear in the table.
Value List	In conjunction with the Input property, defines how the Display List values are identified in the database. Value and Display Lists are entered using the Edit List dialog box. You can enter hard coded entries for each list, or you can supply a variable as the first and only entry. The run time values of the variable are used to populate these lists. This property is supported only in association with the toolbar configuration Knowledge Management Extended, which is not intended to be used outside of the out-of-box Knowledge Management module.

Image File	Specifies the name of an image file to display on the control. The image file must reside in one of the following paths: **Windows client**: • <Service Manager>\Client\plugins\com.hp.ov.sm.client.common_x.xx.xxx\src\resources\icons • A local directory on the client machine, which is specified in the **Image path** field (**Window** > **Preferences**). For example: C:\smCustomImages. **Web client**: • <Service Manager web tier>\images. For example: <Tomcat>\webapps\webtier-x.xx\images. **Note** Note You don't need to include the image file suffix for this property; the image file name must be unique. For example, you would like to display an image instead of text for the Priority column. To do this, follow these steps: 1. Open the probsummary.qbe.g form in the Design view of Forms Designer 2. In the left-hand pane, click the Priority column header. 3. In the right-hand (Table Column) pane, update the Value List option to "1;2;3;4" and the Image File option to "critical;high;mid;low". 4. Add four image files (named "critical.png", "high.png", "mid.png", and "low.png") to the following paths: ○ Windows client: ...\plugins\com.hp.ov.sm.client.common_x.xx\src\resources\icons\obj16 ○ Web client on Tomcat (for example): ...\<Tomcat>\webapps\webtier-x.xx\images\obj16
Display List	Specify the values that appear in the drop-down list at run time. There must be a one–to–one correspondence between the values for Value List and for Display List. If the Display List is the only populated property, the display values are what is written to the database.

Select Only	Setting Select Only to Yes by checking the box requires the user to select from the Combo Box drop-down list. If the user makes a manual entry, the value must already exist in the drop-down list (defined in the Value List and Display List properties). Setting it to No by leaving the box unchecked allows the user to select from the drop-down list or to manually enter a value. No works with or without a drop-down list as defined in the and Display List properties.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.
Maximum Chars	Specify the number of characters the user can enter in the field. The default is 0 (unlimited).
Maximum Characters Beep	When set to Yes by checking the box, it notifies the user by sounding a beep that they reached the maximum number of characters. Not available on UNIX systems.
Case Conversion	Specify how to change the case of text entered in the field. The choices are None, Upper, Lower.
Decimals	Specify the number of decimal points for numbers in this field. Service Manager modifies numbers entered in the field to the number of decimal points you specify.
Parse	Select this option to verify that the text entered in a field is the correct syntax. The syntax for the field is defined by the type of field it displays (date, expression, number, and so forth).
Justification	Specify the justification for the object label as Left, Right, or Center.
Input Conversion	Specify the name of an Input Conversion routine. This property is used by RAD subroutines that mask the display of data on an input field, or check and validate the entry of data into an input field. If specified, the routine is called at runtime to convert the contents of the field before it is displayed on the screen. For more information, refer to RAD input and output conversion.

Mask	Specify the mask value (the condition that sets a value to be passed for this field to the input and output conversion routines). When a field has a mask value and you save the record, the format of the mask value displays in the field. For example, you decide to update the purchase cost information of a contract in Configuration Management. You enter a date in the Currency EX Date field of the Financial tab as 02/26/09. The condition for the mask value is set as: **xxx dd,yyyy**. When you save the record, the **Currency Ex Date** field displays the date format of February 26, 2009. Valid values for the mask vary depending on the routine being called.
Output Conversion	Specify the name of an Output Conversion routine. This property is used by RAD subroutines that mask the display of data on an output field, or check and validate the entry of data into an input field. If specified, the routine is called at runtime to convert the contents of the field before it is displayed on the screen. For more information, refer to RAD input and output conversion.
Password	Select this option to make the text entered in the field a series of asterisks on the screen. The data is not encrypted, just transposed.

1.1.1.2.35. Table control

Use this control to add a table that displays one or more columns of data in a scrollable pane. The tables you create in the Form Designer appear as a rectangular region subdivided by rows and columns. The look and feel mimics tables in Microsoft Office applications, such as Excel.

Table have the following features:

- Cells may display text or images.
- Columns may vary in width. The user can resize the columns using the mouse.
- Columns may have an optional heading button, which can be mapped to perform an action.
- All rows are the same height, defined by the table's font.
- The user can scroll through data by use of scroll bars or the keyboard.
- All expected keyboard navigation is supported (Up, Down, Next, Prev, Last, First, and so forth)
- You can edit cells or make them read-only.
- Selecting a read-only cell results in a highlight of the entire row.
- Selecting an cell to edit results in the in-place creation of an edit field or drop-down Combo Box that lets the user edit the cell.
- Users can edit a field by clicking on it or by pressing the **Enter** key while the table has focus. Tabbing events received while the table is in edit mode result in navigation from one cell to the next. Pressing **Enter** while a cell is being edited results in the changes being accepted and the cell's row being highlighted. Tabbing events received when a row is highlighted are handled by passing focus to the next object in the window.



Note

Note To avoid unpredictable behavior, don't set a DVD condition to use a variable that's based on a record displayed in the Table widget. The only exception is the Foreground Color Condition property, which takes effect at the record level.

To place a table on a form, click **Table** and then click the form.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This field is required if you enable the Multiple Selection property. This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Font	Select a font for the text of your label, table, timer, or marquee caption from the drop-down list.
Foreground Color Condition	Specify a formula to evaluate the ForeColor property at run time. The result of this evaluation overrides the value assigned to the property.
Bold Condition	Specify an expression to override the Bold property when that expression evaluates to true.
Italic Condition	Specify an expression to override the Italic property when that expression evaluates to true.
Font Increase	Specify the amount, in points, to increase or decrease the standard font size.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.

Read-Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to true.
Columns	Specify a list of columns in the table.
Column Headings	Select this option to make the column headings appear above the columns.
Multiple Selection	Select this option to allow the user to select several rows in the table. Clear the checkbox to allow the user to select just one row. **Note** Note • If you enable this option, you must enter the associated table's name in the Name property. • The Multiple Selection property in Forms Designer has no effect on enabling or disabling the multi-select capability when performing a fill from an array field. • The Multiple Selection functionality applies to multiple rows in a single table only. Even if this option is enabled, selecting multiple rows in multiple tables will lead to unexpected results.
Selection Field	Select this option to allow the user to select several rows in the table. Clear the checkbox to allow the user to select just one row.
Default Button ID	When an individual column Button ID is set to 0, the Default Button ID is used.
Double–Click Button ID	Specify the Button ID returned by the double-click action.
Double–Click Field	Specify the field to return to cursor.field.contents() which can then be used in a query to find the record.

Accessible Description	Specify a component description for use with accessibility software (JAWS). (Optional) **Note** Note This function is supported only by the web client. If no description is present, the Web client uses the default table summary as described below: • For a record list on a list page or on the list panel of a list-detail page: **Record List** + **[grouped by]** + **[sorted by]** + **[ascending/descending]**. Example: **Record List Sorted by Update Time Ascending 11x51** • For a Table widget on a detail form: **Table**. Example: **Table 4x3**

Related topics

Form creation
Forms Designer
Creating and editing forms
Using the drawing canvas
Forms Designer properties view
Setting properties
Forms Designer best practices
Building accessible forms
Access Forms Designer
Create a form using the Form Wizard
Update a form
Table column control
Forms Designer controls and tools

1.1.1.2.36. Text control

Use this control to add a text box that displays the contents of a text field and conditionally enables users to enter or modify its contents.

To place a text box on a form, click **Text** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Accessible Name	Specify a component name for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no name is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, it uses the component's text. Accessibility software can query the Web client to get this information and use it to present data to the user in a variety of ways. For example, speech simulation software says the name and the type of component when the component has focus.

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information and use it to present it to the user in a variety of ways. For example, when a component gets focused, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button, or checkbox, you will almost always use the default of 0.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read-only fields a tab–stop value of –1 to prevent users from tabbing into them.
Read-Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to true.
Mandatory	Setting Mandatory to Yes by checking the box indicates that the field is required and therefore a red asterisk indicates a required field is displayed. This is a visible change only. To make the field mandatory for any form it appears on, use the System Definition Utility. To make the field mandatory for a small number of forms only, use Format Control.
Mandatory Condition	Specify an expression to override the Mandatory property when that expression evaluates to true.
Password	Select this option to make the text entered in the field a series of asterisks on the screen. The data is not encrypted, just transposed.
Maximum Chars	Specify the number of characters the user can enter in the field. The default is 0 (unlimited).
Maximum Characters Beep	Select this option to notify the user by sounding a beep that they reached the maximum number of characters. This is unavailable on UNIX systems.
Case Conversion	Specify how to change the case of text entered in the field. The choices are None, Upper, Lower.

Decimals	Specify the number of decimal points for numbers in this field. Service Manager modifies numbers entered in the field to the number of decimal points you specify.
Parse	Select this option to verify that the text entered in a field is the correct syntax. The syntax for the field is defined by the type of field it displays (date, expression, number, and so forth).
Array Length	Specify the size of the scrolling region used to view array entries. A scroll bar appears beside the fields to allow users to view the array entries. - If a field is assigned an Array Length of 5, the form stacks five fields vertically to allow users to view the five array entries. - If Array Length is set to -1, the form dynamically creates as many fields as there are entries in the array. - If the Input data type is scalar, only a single text box appears. **Note**: The screen object must be associated with an array data structure. The default is 0, which means one vertical line of information appears.
Input Conversion	Specify the name of an Input Conversion routine. This property is used by RAD subroutines that mask the display of data on an input field or check and validate the entry of data into an input field. If specified, the routine is called at runtime to convert the contents of the field before it is displayed on the screen. For more information, refer to RAD input and output conversion.
Mask	Specify the mask value (the condition that sets a value to be passed for this field to the input and output conversion routines). When a field has a mask value and you save the record, the format of the mask value displays in the field. For example, you decide to update the purchase cost information of a contract in Configuration Management. You enter a date in the Currency EX Date field of the Financial tab as 02/26/09. The condition for the mask value is set as **xxx dd,yyyy**. When you save the record, the **Currency Ex Date** field displays the date format of February 26, 2009. Valid values for the mask vary depending on the routine being called.

Output Conversion	Specify the name of an Output Conversion routine. This property is used by RAD subroutines that mask the display of data on an output field, or check and validate the entry of data into an input field. If specified, the routine is called at runtime to convert the contents of the field before it is displayed on the screen. For more information, refer to RAD input and output conversion.
Placeholder	Specify the placeholder that is displayed in the text control.

1.1.1.2.37. Text area control

Use this control to add a text area that displays the contents of a text field and conditionally enables users to input several lines of data. This object contains scroll bars and allows text wrapping.



Note

Note A field associated to a text area must be defined as an array of characters in the dbdict.

To place a text area on a form, click **Text area** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Using URLs in text areas

Text areas (including those in dynamic forms) support clickable URLs. When a text area is not in focus (that is, when it is in view mode), URLs in the text area are displayed as hyperlinks. Clicking the hyperlink opens the URL in a new page of the browser. When a text area is in focus (that is, when it is in edit mode), you can modify the text in the text area, but the URL is not displayed as a hyperlink and is not clickable.

The following URL protocols are clickable when used in text areas:

- http://
- https://
- ftp://
- notes://

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Accessible Name	Specify a component name for use with accessibility software. (Optional) **Note**:This function is only supported by the web client. If no name is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, it uses the component's text. Accessibility software can query the Web client to get this information and use it to present data to the user in a variety of ways. For example, speech simulation software says the name and the type of component when the component has focus.

Accessible Description	Specify a component description for use with accessibility software. (Optional) **Note**: This function is only supported by the Web client. If no description is present, the Web client defaults to looking for a nearby label to use for the name of the component. For buttons, checkboxes, or radio buttons, the Web client will use the component's text. Accessibility software can query the Web client to get this information, and use it to present it to the user in a variety of ways. For example, when a component gets focus, speech simulation software will say the name and the type of component, based on the settings in the Accessible Name and Accessible Description fields.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button or checkbox, you will almost always use the default of 0.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.
Read-Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to true.
Mandatory	Select this option so that a red asterisk indicating a required field is displayed. This is a visible change only. - To make the field mandatory for any form it appears on, use the System Definition Utility. - To make the field mandatory for a small number of forms only, use Format Control.
Mandatory Condition	Specify an expression to override the Mandatory property when that expression evaluates to true.
Password	Select this option to make the text entered in the field a series of asterisks on the screen. The data is not encrypted, just transposed.
Maximum Chars	Specify the number of characters the user can enter in the field. The default is 0 (unlimited).
Maximum Characters Beep	Select this option to notify the user by sounding a beep that they reached the maximum number of characters. This is unavailable on UNIX systems.
Case Conversion	Specify how to change the case of text entered in the field. The choices are None, Upper, Lower.

Decimals	Specify the number of decimal points for numbers in this field. Service Manager modifies numbers entered in the field to the number of decimal points you specify.
Parse	Select this option to verify that the text entered in a field is the correct syntax. The syntax for the field is defined by the type of field it displays (date, expression, number, and so forth).
Extend Enabled	Select this option to display two buttons (Expand and Restore) above this control in the web client. Each time you click the **Expand** button, the height of the text area is increased by 40px. After clicking the **Restore** button, the height is restored to the initial value.

1.1.1.2.38. Tag control

Use this control to add a text box that displays the contents of a text field and conditionally enables users to enter one or more tags.

To place a text box on a form, click **Tag** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	(Optional) Specify a unique identifier for the object on the screen. This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to `true`.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Input	Specify the database field or variable to associate with this control.
Tab Stop	Specify the tab sort order for this field. By default, the TAB key navigates the focus on a form from top to bottom and left to right. Objects that have a non-zero Tab Stop are visited first, in ascending order. For a button, file, radio button, or checkbox, you will almost always use the default of 0.
Read-Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read-only fields a tab-stop value of -1 to prevent users from tabbing into them.
Read-Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to `true`.
Mandatory	Setting Mandatory to Yes by checking the box indicates that the field is required and therefore a red asterisk indicating a required field is displayed. This is a visible change only. To make the field mandatory for any form it appears on, use the System Definition Utility. To make the field mandatory for a small number of forms only, use Format Control.
Mandatory Condition	Specify an expression to override the Mandatory property when that expression evaluates to `true`.

1.1.1.2.39. Timer control

Use this control to add a clock timer to the form. The timer initiates events at a certain time, based on settings in the Input and Expiration Event fields.

To place a clock timer on a form, click **Timer** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Font	Select a font for the text of your label, table, timer, or marquee caption from the drop-down list.
Foreground Color	Select the text color from the drop-down list.
Foreground Color Condition	Specify a formula to evaluate the ForeColor property at run time. The result of this evaluation overrides the value assigned to the property.
Background Color	Specify the Background Color. You must select the Opaque property checkbox for the Back Color to have an effect. The only Background Color available for a Windows client marquee is black.
Background Color Condition	Specify an expression to override the BackColor property when that expression evaluates to true.
Bold	Select this option to apply bold to the label, default text, frame caption, or marquee caption.
Bold Condition	Specify an expression to override the Bold property when that expression evaluates to true.

Italic	Select this option to apply italics to your label, default font, or caption.
Italic Condition	Specify an expression to override the Italic property when that expression evaluates to true.
Font Increase	Specify the amount, in points, to increase or decrease the standard font size.
Font Increase Condition	Specify an expression to override the Font Increase property when that expression evaluates to true.
Output Conversion	Specify the name of an Output Conversion routine. If specified, the routine is called at runtime to convert the contents of the field before it is displayed on the screen. For more information, refer to RAD input and output conversion.
Array Length	Specify the size of the scrolling region used to view array entries. A scroll bar appears beside the fields to allow users to view the array entries. - If a field is assigned an Array Length of 5, the form stacks five fields vertically to allow users to view the five array entries. - If Array Length is set to -1, the form dynamically creates as many fields as there are entries in the array. - If the Input data type is scalar, only a single text box appears. **Note**: The screen object must be associated with an array data structure. The default is 0, which means one vertical line of information appears.
Expiration Color	Specify the color that the timer should change to when it reaches zero.
Expiration Event	Specify the event to send to the server when the timer reaches zero. Specify a numeric value; for example, a button or event ID.

Related topics

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Creating and editing forms
Using the drawing canvas
Forms Designer properties view
Setting properties
Forms Designer best practices
Building accessible forms
Access Forms Designer
Create a form using the Form Wizard

Update a form
Date control
Decimal control
Forms Designer controls and tools

1.1.1.2.40. Visualization control

Use this control to graphically display Configuration Item (CI) relationships.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Read–Only	Select this option to disable editing capabilities and provide only viewing access to the field. Give read–only fields a tab–stop value of –1 to prevent users from tabbing into them.
Read–Only Condition	Specify an expression to override the Read-Only property when that expression evaluates to true.

Related topics

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1.1.1.2.41. Web Preview tool

This button enables you to see a preview of what the form will look like when displayed by the Web client. The preview doesn't take into account any active DVD on the form.

Click **Web Preview** to display the form in the Web Preview tab.

Related topics

Form creation
Forms Designer
Creating and editing forms
Using the drawing canvas
Forms Designer properties view
Setting properties
Forms Designer best practices
Building accessible forms
Access Forms Designer
Create a form using the Form Wizard
Update a form
Forms Designer controls and tools
Grid view tool
Selection tool

1.1.1.2.42. Wrap label control

Use this control to add a multi–line label. You can use this label to give a title to the form, give labels to objects within the form, or otherwise place text on the form.

To place a wrapping label on a form, click **Wrap label** and then click the form.
Be sure to put the name of the database field or variable that you want to associate with this control into the **Input** property.

Property	Usage
Name	Specify a unique identifier for the object on the screen. (Optional) This name is used by external applications, such as RAD, to dynamically change the properties of the object.
X	Specify the object's horizontal position based on the left edge of the object.
Y	Specify the object's vertical position based on the top edge of the object.
Width	Specify the width of the object in alignment grid units.
Height	Specify the height of the object in alignment grid units.
Visible	Select this option to make the object visible on the form. Clear the checkbox to hide the object from view on the form.
Visible Condition	Specify an expression to override the Visible property when that expression evaluates to true.
Caption	Specify a text caption for the object.
Caption Condition	Specify an expression to override the Caption property when that expression evaluates to true.
Input	Specify the database field or variable to associate with this control.
Label for	Specify the name of a control that uses this label. **Note** Note If you specify a variable for this label, you need to provide a valid value here. Otherwise, the label can't be read by JAWS.
Foreground Color	Select the text color from the drop-down list.
Foreground Color Condition	Specify a formula to evaluate the ForeColor property at run time. The result of this evaluation overrides the value assigned to the property.
Background Color	Specify the Background Color. You must select the Opaque property checkbox for the Back Color to have an effect. The only Background Color available for a Windows client marquee is black.
Background Color Condition	Specify an expression to override the BackColor property when that expression evaluates to true.

Font	Select a font for the text of your label, table, timer, or marquee caption from the drop-down list.
Bold	Select this option to apply bold to the label, default text, frame caption, or marquee caption.
Bold Condition	Specify an expression to override the Bold property when that expression evaluates to true.
Italic	Select this option to apply italics to your label, default font, or caption.
Italic Condition	Specify an expression to override the Italic property when that expression evaluates to true.
Font Increase	Specify the amount, in points, to increase or decrease the standard font size.
Font Increase Condition	Specify an expression to override the Font Increase property when that expression evaluates to true.
Array Length	Specify the size of the scrolling region used to view array entries. A scroll bar appears beside the fields to allow users to view the array entries. • If a field is assigned an Array Length of 5, the form stacks five fields vertically to allow users to view the five array entries. • If Array Length is set to -1, the form dynamically creates as many fields as there are entries in the array. • If the Input data type is scalar, only a single text box appears. The default value is 0, which means one vertical line of information appears. **Note** Note The screen object must be associated with an array data structure.
Heading Level	Specify the heading level for this control so that a corresponding HTML heading tag (<h1> to <h6>) is generated. **Note** Note This function is only supported by the Web client.

Related topics

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Label control
Link label control
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1.1.1.3. Forms Designer best practices

Forms Designer can produce portable forms that render successfully in the Windows and Web clients if you follow a few design suggestions. To ensure all forms are portable, test each new or upgraded form with both clients.

Hint: You can cut or copy and paste existing objects onto other forms.

Printing forms: printing is designed to work nicely with label/inputs; however, it will NOT work with arbitrary positioning of widgets in the form. If you want to print a table, make it a table - don't simulate one with "cleverly" placed labels.

Using pop-up subforms can save screen real estate, and can be helpful for performance because the query to obtain the subform information is not made until the data is needed for display.

Form design

The right edge of any form should be at grid unit 156. The bottom of any form should be at grid unit 42. Forms which follow this guideline will fit in the default fonts at 640x480 and 800x600. Additionally, these forms provide enough room along the edges for a scroll bar if required.

When placing controls on forms, make all character dimensions (coordinates for placement on the screen, height, width) even numbers, because the space required to display one character is 2x2. When the ArrayLength of a field is set to be a number greater than 0 and the field is situated at an odd coordinate, it will automatically shift to an even coordinate.

Important: Each field needs a unique name property. You can change the value in the name property, but don't delete it.

Things to keep in mind when designing forms.

- Provide logical navigation buttons in all forms. For example, the Back, Cancel, Submit buttons.
- Use a Submit or Search button where appropriate instead on relying solely on the ENTER key.
- Stack wide elements like text areas used for descriptions. Multiple wide fields can make a form to wide. Stacking is easier to read than wide elements placed side-by-side.
- Avoid creating wide forms. Recent surveys show that the majority of users don't use monitor settings greater than 1024x768.
- Avoid three–column layouts. Scrolling vertically is more acceptable to most people that scrolling horizontally.
- Avoid adding too many fields, widgets, and notebook tabs to a single form. Simplify forms by breaking it up into smaller forms that link to other forms through a logical flow. Too much information is overwhelming to most users and results in a less productive application. Too many controls on a form can severely degrade performance on the Web client.
- Avoid placing elements too close to each other. Forms are easier to read and use when the text and fields are spaced out. When users increase font sizes this becomes more apparent. This can translate into overlapping form elements, text, and containers in the Web and Windows clients.
- Printing works well with label/inputs. It does NOT work with arbitrary positioning of widgets in the form. If you want to print a table, make it a table. Do not simulate one with cleverly placed labels.
- When you design a QBE list form that contains a table, don't enclose the table in a Frame or Group; otherwise the QBE list form will not function properly.

Form layouts

Follow these guidelines to generate forms that display well on a variety of platforms and with different screen resolutions and fonts.

The size of Service Manager forms and the objects on them are defined in terms of grid units. For instance, a combo box may be defined to be 36 units wide and 2 units tall. Objects are also placed on forms using the same units. For example, the same combo box may have its upper left corner at a location 5 units to the right of the form edge and 4 units beneath the top of the form.

The size of a grid unit varies depending upon the currently selected Service Manager font. The grid unit is always defined as being half as wide as the lower case *e* in the current font, and half as tall as the lower case *e* in the current font. Thus, in a font whose letter *e* is 8 pixels wide by 12 pixels tall, the grid unit is 4 pixels wide and 6 pixels tall.

Changing the system's font changes the size of a form on the screen. Just as importantly, it may change the relative shape of the objects therein. If a screen's grid unit goes from 4x4 to 6x8, then objects on the screen becomes 50% taller, and doubles in width. Thus, the screen appears to stretch and have different proportions than it did originally.

An important point to recognize is that Windows true-type fonts are non-deterministic. Each video driver manufacturer can bundle its own hardware mapping of common fonts, and most modern video cards do so. Many manufacturers improve upon the base Microsoft Windows definition of what constitutes a particular font. Thus, the letter *e* in Arial 8 pt. bold when displayed at 640x480 on one video card may have a different metric than the same letter displayed at the same resolution on a different video card.

Use easy to understand names and descriptions for buttons and links. Acronyms and cryptic language makes it is difficult or impossible to know what's going to happen if you click the link or press the button.

Fonts

Using font styles such as Bold or Italic affects the spacing of the characters on a form. If labels are truncated in forms, select a different font, style, or size. You can also increase the width of the form to accommodate a different font, style, or size by specifying an increased width percentage in Service Manager preferences.

Web client forms

The Service Manager Web tier offers automatic support for existing applications and their forms. The Web tier generates dynamic HTML that approximates the exact layout of forms as you define them with Forms Designer. The default application forms are portable from the Windows client to the Web client with no required modification.

If you upgrade an existing system, you will find that Service Manager clients automatically support and display your customized forms. However, there may be cases when further form modification is necessary to correct cosmetic issues that appear when you view the form with the Web client. The following table describes common form revisions that might be required.

Area	Correction
Overlapped objects	Ensure that form objects don't overlap each other. The Windows client makes slight adjustments to correct for overlapped objects, but these design issues are exposed on a web browser.
Dynamic resizing	The Windows client dynamically resizes objects such as text areas and notebooks when you resize the window. The Web client doesn't resize most objects and doesn't support elastic properties as you resize the browser window. Therefore, ensure that you assign a default initial size.
Graphics and images	Size any graphics and images to fit conservatively in the form. The Windows client supports scaled images used as buttons. However, the Web client displays these images in their native size. The button grows to accommodate the size of the image.
Dynamic View Dependency (DVD) conditions	Hidden data with DVD conditions may be exposed if the form permits the user to make changes. **Note** Note If you are using visible DVD and the system has dynamically changed the read-only attribute of a UI component by using JavaScript, the value of this component is still possible to be changed. For sensitive field data, administrators should consider the use of Data Policy expressions instead of DVD or a combination of the two to ensure data integrity. Don't rely on DVD alone to ensure data integrity for sensitive field data as it may be changed by users familiar with browser developer tools.

Collapsible sections

In the Web client, groups of items can be displayed in collapsible sections. To display such items, use the Group control in Forms Designer, and check the Collapse Enabled property.

When designing forms with collapsible sections, use the following guidelines to avoid formatting problems when the form is displayed in the Web client.

Important guidelines

- All groups on the form should be marked as Collapsible-enabled.
- All widgets must be placed in a collapsible group box; no widgets should remain floating independently on the form.
- Do not stack or mix collapsible group boxes with non-collapsible group boxes.

Sizing graphics

When creating or placing images in the Service Manager GUI, create the image in the size needed. Do not create oversized or undersized images and expect Service Manager to resize them properly. There are several important reasons for this.

- When scaling images, the quality of the image distorts.
- The size of the image is not changed and thus may be a performance consideration.
- The Web client has no way of resizing the image. For example, in the Service Manager GUI, if you placed a large image in a button, the GUI and the Windows client size it properly based on the button container. The Web client doesn't do this and has no way of scaling the image to the size of the parent container. Thus the image appears as its original size in the Web client. The bigger the original size, the bigger it appears in the Web client. On the Web this can also be a major performance issue because large images require more bandwidth and cause slow browser performance.

Form naming conventions

Form names incorporate the application or feature identification as a prefix in the form name. Look for these prefixes to identify updated or new forms.

Application	Form prefix	Related table
Problem Management	PM	knownerror rootcause rootcausetasks
Service Level Management	sla	sla
Service Desk	SM	incidents
Incident Management	IM	probsummary

1.1.1.4. Building accessible forms

You can use the Forms Designer to make your system more accessible. In addition to the accessible attributes, you can customize forms for easier readability and information manipulation.

Attribute name	Definition
accessiblename option	Give a form element a name compatible with a screen reader.
accessibledescription option	Give a form element a description compatible with a screen reader.

Accessible Web client forms[edit]

The accessible mode of the Web client allows users that require different levels of accessibility to apply personal preferences to improve their user experience. The accessible Web client also enables accessibility tools, such as screen readers, to work with Service Manager. The accessible Web client omits the graphical workflow feature, and thread navigation links, which are the tabs that identify open forms in the Windows client.

If you are designing forms for accessible use, these are the most important design requirements:

- High-contrast color graphics
- Larger default fonts
- Larger form spacing (more white space)
- Simplified navigation (fewer buttons, objects and icons)
- Browser settings must be able to control

- Resizing fonts
- Foreground and background color selection

A visually impaired user might want a well–designed form that reads left to right with labels announcing the name of the subsequent form object and tables with row labels that read horizontally (not by column). This user might also want to specify a black background with white text in a 14–point bold font, instead of the default color and font combinations.

Putting the HTML Editor on accessible forms[edit]

The HTML Editor must be the last widget in its form for reverse tabbing to work properly. Widgets placed under it can't be tabbed.

The Accessible Web client ignores the tab stop information set in the Form definition. Instead, the tabbing always moves from left to right and top to bottom so that the tabbing order matches the order in which the widgets are described by a screen reader.

For complex editing requiring 508 compliance, edit the text in a 508 compliant HTML Editor and then paste it into the HTML Editor.

Contents

- 1 Accessible Web client forms
  - 1.1 Putting the HTML Editor on accessible forms

1.1.2. Dynamic View Dependencies

Steps taken to ensure better data entry and form usability are critical to a database. This includes visual enhancements and form modifications based on user input. Dynamic View Dependencies (DVD) is a feature of Forms Designer that enables you to display objects and their attributes on a form depending on user interaction.

DVD enables you to create control or layout objects with properties that are evaluated when the form appears to control the appearance of these objects on the form.

You can use dynamic view dependencies to:

- Dynamically determine the properties of a field based on the DVD condition.
- Make a field visible only when another field contains a certain value.
- Dynamically determine the choices in a drop-down list based on the choice in another combo box.
- Dynamically set the font or color of a field based on the value of another field.
- Dynamically set the font or color of a field based on its value.

Note: DVD select statements can only be used on text fields.

You create dynamic view dependencies by specifying conditional statements for properties of a control. The three types of conditional statements for a property are described in the following table.

Conditional Statements	Syntax	Example
Simple expression (**Field comparisons**): • Enable you to specify comparisons of equality or inequality. • Evaluate to "true" or "false" at runtime.	FIELD [OPERATOR VALUE] • FIELD: Any valid input field or variable enclosed in brackets. • OPERATOR: `=` , `>` , `<` , `<>` , or `#` • VALUE: Any quoted string or number **Note**: You can use `|` or `&` to concatenate multiple expressions. For more information, see Field comparisons.	• [$invisible] = 0 • [$invisible] <> 1 • [$L.one.click]<>true & [$quantity]>1& [$orderonbehalf.marker] <>true
Value:Display expression (**Field value matching**) • Enable you to specify matching output values for various input values. • Return a single value or a list of values when evaluated at runtime.	FIELD ? ValueList : DisplayList • FIELD: Any valid input field or variable enclosed in brackets. • ValueList and DisplayList: Zero or more comma separated values. For more information, see Field value matching.	[$severity]? "sev1","sev2","sev3" : 1,7,2,13 This example returns a value as follows: • 1 if $severity="sev1" • 7 if $severity="sev2" • 2 if $severity="sev3" • 13 if $severity is none of the listed "sev" values
Function call (**Dynamic functions**): • Enable you to call a function that evaluates a result dynamically.	FUNCTION(ParaList) • FUNCTION: A valid function name (for example, `select`) • ParaList: Zero or more comma separated VALUEs For more information, see Dynamic functions.	select("subcategory", "subcategory", "category", [$category]) This example returns a list of subcategory names from the `subcategory` table, whose category equals the `$category` value.

Service Manager includes a form, **dep.g**, which demonstrates Dynamic View Dependencies.

1.1.2.1. Use form dep.g to demonstrate Dynamic View Dependencies

The form dep.g has been prepared to demonstrate Dynamic View Dependencies. You can interact with this form to see the DVD examples this form uses and the dynamic effects they create.

To understand DVD examples on the dep.g form:

1. Click **Tailoring** > **Forms Designer**.
2. Type `dep.g` in the Form field, and click **Search**.
   The `dep.g` form opens.
3. Select the **Instructions** tab for instructions on how to use three types of DVD conditional statements. For more information, see Dynamic View Dependencies.
4. Follow the instructions and interact with the different fields to observe how the View properties change dynamically.
5. Click **Design**.
6. Inspect Properties View to see what values are assigned to create the effects. The following tables explain those properties that create the dynamic effects on different tabs of the form.

Labels and Fields tab:

Control Name	Property: Value	Explanation
Providers > Caption text box (Text22)	○ Visible: 1 ○ Input: $caption	○ This text box is always visible. ○ The value you type in this text box (**disabled**, **big**, **bold**, or **italic**) is assigned to the $caption variable, which is referenced in other conditional statements on the same screen.
Providers > Severity text box (Text25)	○ Visible: 1 ○ Input: $severity	○ This text box is always visible. ○ The value you type in this text box (**sev1**, **sev2**, **sev3**, or any other value) is assigned to the $severity variable, which is referenced in other conditional statements on the same screen.
Providers > Disabled checkbox (CheckBox28)	○ Visible: 1 ○ Input: $disabled	○ This checkbox is always visible. ○ When you select or clear this checkbox, 1 or 0 is assigned to the $disabled variable, which is referenced in other conditional statements on the same screen.
Providers > Invisible checkbox (CheckBox30)	○ Visible: 1 ○ Input: $invisible	○ This checkbox is always visible. ○ When you select or clear this checkbox, 1 or 0 is assigned to the $invisible variable, which is referenced in other conditional statements on the same screen.

Dependents > Caption label (Label33)	○ Visible Condition: [$invisible] = 0 ○ Caption Condition: [$caption] ○ Foreground Color Condition: [$severity]	○ This label is visible only when the Providers > Disabled checkbox (CheckBox28) is not selected. ○ The value you type in the Providers > Caption text box (Text22) is automatically populated as the caption of this label. ○ The font color of the label caption changes with a color ID number (1, 2, 3 and so on) you type in the Providers > Severity text box.
Dependents > Expression label (Label35)	○ Visible Condition: [$invisible] = 0 ○ Foreground Color Condition: [$severity] ? "sev1","sev2","sev3" : 1, 7, 2, 13 ○ Bold Condition: [$caption]="bold" ○ Italic Condition: [$caption]="italic" ○ Font Increase Condition: [$caption] ? "big" : 5, 0	○ This label is visible only when the Providers > Disabled checkbox (CheckBox28) is not selected. ○ The font color of the label caption changes with the value you type in the Providers > Severity text box (Text25): color 1 for sev1, color 7 for sev2, color 2 for sev3, and color 13 for any other entries. ○ The label caption is bolded if you type **bold** in the Providers > Caption text box (Text22). ○ The label caption is in italic if you type **italic** in the Providers > Caption text box (Text22). ○ The font size of the label caption is increased by 5 only when you type **big** in the Providers > Caption text box (Text22).

Dependents > Disable Checkbox text box (Text37)	○ Visible Condition: [$invisible] = 0 ○ Read-Only Condition: [$disabled] ○ Input: $x	○ This text box is visible only when the Providers > Disabled checkbox (CheckBox28) is not selected. ○ This text box is read-only when the Providers > Disabled checkbox (CheckBox28) is selected. ○ The value you type in this text box is assigned to the $x variable, which is also the Input of the Disable Caption text box (see below).
Dependents > Disable Caption text box (Text39)	○ Visible Condition: [$invisible] <> 1 ○ Input: $x ○ Read-Only Condition: [$caption] = "disabled"	○ This text box is visible only when the Providers > Disabled checkbox (CheckBox28) is not selected. ○ The input of this text box is $x, which is same as the value in the Dependents > Disable Checkbox text box (Text37). ○ This text box becomes read-only when you type **disabled** in the Providers > Caption text box (Text22).
Dependents > Visible label (Label41)	○ Visible: 1 ○ Caption Condition: [$invisible] ? 0, 1 : "Yes" ,"No"	○ This label is always visible. ○ The caption of this label is **Yes** when the Providers > Disabled checkbox (CheckBox28) is not selected, and **No** when the checkbox is selected.

Select Function tab:

Control Name	Property: Value	Explanation
Categories combo box (ComboBox51)	◦ Input: $category ◦ Value List Condition: select("name", "category", "category", *)	◦ The value you select in this combo box is assigned to the $category variable, which is referenced in the Value List Condition property of the Subcategories combo box. ◦ This select function returns all category names from the `category` table (that is, the `category` field can be any value (*)).
Subcategories combo box (ComboBox53)	◦ Input: $x ◦ Value List Condition: select("subcategory", "subcategory", "category", [$category])	◦ The value you select in this combo box is assigned to the $x variable (which is not referenced in any other conditional statements on the same screen though). ◦ This select function returns, from the `subcategory` table, the subcategories of a category you specify in the Category combo box ($category).
Operator combo box (ComboBox57)	◦ Input: $operator ◦ Value List Condition: select("name", "operator", "name", *)	◦ The value you select in this combo box is assigned to the $operator variable. ◦ This select function returns the login names of all records in the `operator` table (that is, the login name can be any value (*)).
Capabilities text area (MultiText59)	◦ Caption Condition: select("cap.exec", "operator", "name", [$operator])	This select function returns the capability words (the cap.exec field in the `operator` table) of the operator that you select in the Operator combo box into the Capabilities text area.

Buttons tab:

Control Name	Property: Value	Explanation
Ok radio button (Radio70)	○ Input: $buttonID ○ Value: Ok	○ The input value of this radio button is assigned to the $buttonID variable. ○ When selected, this radio button has an input value of **Ok**.
Cancel radio button (Radio72)	○ Input: $buttonID ○ Value: Cancel	○ The input value of this radio button is assigned to the $buttonID variable. ○ When selected, this radio button has an input value of **Cancel**.
Design radio button (Radio73)	○ Input: $buttonID ○ Value: Design	○ The input value of this radio button is assigned to the $buttonID variable. ○ When selected, this radio button has an input value of **Design**.
None radio button (Radio74)	○ Input: $buttonID ○ Value: None	○ The input value of this radio button is assigned to the $buttonID variable. ○ When selected, this radio button has an input value of **None**.

Button (Button71)	○ Caption Condition: [$buttonID] ○ Button ID Condition: [$buttonID] ? "Ok", "Cancel", "Delete", "Design" : 3, 5, 4, 9, 50	○ The caption of this button dynamically changes to the value of the $buttonID variable (Ok, Cancel, Design, or None). ○ The ID of this button changes dynamically with the $buttonID variable value as follows: ■ 3 when the $buttonID value is Ok; ■ 5 when the $buttonID value is Cancel; ■ 4 when the $buttonID value is Delete; ■ 9 when the $buttonID value is Design; ■ 50 when the $buttonID value is none of the above-listed values When you press the button, it functions as the Ok, Cancel, or Design button accordingly.

Bitmaps tab:

Control Name	Property: Value	Explanation
Provider > Ticket Severity combo box (ComboBox79)	○ Input: $ticketSeverity ○ Value List: ok;warning;alert ○ Display List: Low;Medium;High	○ The value you select in this combo box is assigned to the $ticketSeverity variable. ○ The available selections are: ok, warning, and alert. ○ The available selections are displayed as: Low, Medium, and High.

Dependents > chart (Chart81)	○ Caption Condition: [$ticketSeverity] ? "ok", "warning", "alert" : "1","5","10","0" ○ Color List=green;yellow;red ○ Color Scale=1;5;10	**Note**: The Properties pane of this chart doesn't currently display the complete property information. You can open the dep.g form in non-Design mode and click **Format Detail** from the More Actions menu to view the complete property information of this chart. ○ A value displays in the column of the chart as follows: ▪ 1 when $ticketSeverity is "ok" (that is, when you select **Low** in the Ticket Severity combo box); ▪ 5 when $ticketSeverity is "warning" (that is, when you select **Medium** in the Ticket Severity combo box); ▪ 10 when $ticketSeverity is "alert" (that is, when you select **High** in the Ticket Severity combo box); ▪ 0 when $ticketSeverity is none of the above-listed values (that is, when the Ticket Severity combo box is empty). ○ A list of colors is applied to the chart column: green, yellow, and red. ○ The color of the chart column is determined as follows: ▪ Green when the column value is 1; ▪ Yellow when the column value is 5; ▪ Red when the column value is 10.
Dependents > image (Graph82)	○ Image Condition: [$ticketSeverity]	This image dynamically changes with the option you select in the Ticket Severity combo box: Low (ok.gif), Medium (warning.gif), or High (alert.gif).

Dependents > marquee (Marquee83)	○ Caption Condition: [$ticketSeverity] ? "ok","warning","alert" : "Severity is low ...","Severity is medium ...","Severity is HIGH!!"	The caption of this marquee dynamically changes with the option you select in the Ticket Severity combo box: ○ **Severity is low ...** when you select **Low** (ok); ○ **Severity is medium ...**when you select **Medium** (warning); ○ **Severity is HIGH!!** when you select **High** (alert).

Circular Dependencies tab

Control Name	Property: Value	Explanation
Hello checkbox (CheckBox88)	○ Input: $hello ○ Selected Condition: [$world]	○ The input value of this checkbox (0 or 1) is assigned to the $hello variable. ○ This checkbox is selected or deselected when the value of the $world variable is 1 or 0 (that is, when the World checkbox is selected or deselected).
World checkbox (CheckBox89)	○ Input: $world ○ Selected Condition: [$hello]	○ The input value of this checkbox (0 or 1) is assigned to the $world variable. ○ This checkbox is selected or deselected when the value of the $hello variable is 1 or 0 (that is, when the Hello checkbox is selected or deselected).
I am my number one fan! checkbox (CheckBox91)	○ Input: $self ○ Selected Condition: [$self]	○ The input value of this checkbox (0 or 1) is assigned to the $self variable. ○ This checkbox is selected or deselected when the value of the $self variable is 1 or 0 (that is, select or deselect of this checkbox is independent of any other controls).

Note: The fields in dep.g use variables because dep.g is not associated to a dbdict. You can use field names on forms that do have an associated dbdict.

1.1.2.2. Setting Dynamic View Dependencies properties

You can set up a number of properties for dynamic evaluation using the Properties window of Forms Designer. For example, the Caption property that you set for a Label specifies the text to display on the screen. Under the “Caption” property there is a property named “Caption Condition.” This associated property enables you to specify a condition evaluated dynamically to compute a caption for the label. These two properties are paired for the purpose of DVD conditions. There are several other property pairings as well.

Several Forms Designer controls have “property” and “property condition” pairs. In each case, you can specify a special conditional statement that specifies the run-time value for the property.

DVD conditional property pairings and examples of their supported types of controls are listed in the following table.

Property	Property Condition	Examples of supported control types
Caption	Caption Condition	Button, Check Box, Combo Box, Label, Marquee, Radio Button, Text, and Text Area
Value List	Value List Condition	Combo Box
Display List	Display List Condition	Combo Box
Visible	Visible Condition	Button, Check Box, Combo Box, Label, Marquee, Radio Button, Text, and Text Area
Foreground Color	Foreground Color Condition	Label
Background Color	Background Color Condition	Label
Bold	Bold Condition	Label
Italic	Italic Condition	Label
Font Increase	Font Increase Condition	Label
Read-Only	Read-Only Condition	Check Box, Combo Box, Radio Button, Text, and Text Area
Button ID	Button ID Condition	Button
Image File	Image Condition	Button
-	Selected Condition	Check Box
Mandatory	Mandatory Condition	Combo Box, Radio Button, Text, and Text Area
Enabled	Enabled Condition	Button

1.1.2.3. Using Dynamic View Dependencies vs. Data Policy

Overuse of DVD on forms can result in performance problems. For example:

- Using the select function or a $ variable that's not on the form requires an interaction with the server. On connections with slow network response the use of these types of DVD statements can significantly impact performance.
- Using Focus In/Out or Data Changed Events also causes a server interaction. During design, the response time may seem adequate. However on a network with a half second response, it can be cumbersome.

Data Policy operates at the table level and achieves many of the same results without the complexity and without taxing system resources. For example, one common use of DVD is to set Visible or Readonly conditions, both of which can be set using Data Policy. With Data Policy, the conditions are evaluated on the server before the form is sent, thus avoiding a network interaction.

1.1.2.4. Field comparisons

Field comparisons use a statement to create a conditional property for a control on a form.

Syntax: `FIELD OPERATOR VALUE`

- *FIELD* is any variable or field name surrounded by brackets, for example, `[$x], [name],` and so forth.
- *OPERATOR* can be `=` , `>` , `<` , `<>` , or `#` .
- *VALUE* is any quoted string or number.

Note: Field comparisons can also use complex conditions or statements to create a conditional property for a control on a form. You can use `|` or `&` to concatenate multiple expressions. For example: `[$L.one.click]<>true &[$quantity]>1&[$orderonbehalf.marker]<>true` .

Important: In order for a property to depend on a variable (or record field), the variable must be assigned as an input to an object on the same screen. For examples, see Use form dep.g to demonstrate Dynamic View Dependencies.

Here are some examples of using field comparisons.

- Making a certain field visible only when the category of a record equals "network."
  - Assume that the category can be selected via a drop-down list whose Input is defined as `$category` .
    You can specify the visibility dependency by assigning the following to the Visible Condition property of the field:
    `[$category] = "network"`
  - Instead of a variable, a drop-down list can use a record field as input. If the field is named "category," set up the "Visible Condition" as follows:
    `[category] = "network"`
- Adding a DVD visible condition to a field:

`[assignment]#"help"`

- Specifying an inequality comparison:

`[$category] <> "network"` (that is, category is not network)

- Using complex conditions or statements to create a conditional property for a control on a form.
  - `[$category]="network"&[$subcategory]="modem"`
  - `[$category]="network"|[$category]="DEFAULT"`

Complex conditions are true only if both parts of the condition are true.

1.1.2.5. Field value matching

Field value matching enables you to specify matching output values to various input values.

Syntax: `FIELD ? ValueList : DisplayList`

FIELD is any variable or file field surrounded by brackets, for example, `[$x], [name]` , and so forth.

ValueList and *DisplayList* are lists of comma-separated values

Use field value matching[edit]

Setting the color of a label to change based on the value of a severity field

The severity is specified in an edit field whose input is *$severity*. You set the Color Condition of the label as follows:

```
[$severity]? "sev1","sev2","sev3" : 1,7,2
```

This statement tells the label to do the following:

- Use a color value of 1 (red) when $severity="sev1"
- Use a color value of 7 (yellow) when $severity="sev2"
- Use a color value of 2 (green) when $severity="sev3"

Specify a value to use if no match is found

```
[$severity]? "sev1","sev2","sev3" : 1,7,2, 0
```

This tells the system to use color 0 (black) if *$severity* is none of the listed "sev" values.

Note: The *$severity* field must have been assigned as input to at least one field on the screen.

Contents

- 1 Use field value matching

1.1.2.6. Dynamic functions

The dynamic property condition statement enables you to call a function that evaluates a result dynamically.

Syntax: `FunctionName( PARM_0, ..., PARM_n )`

The parameters for the function can be any string or number. One parameter may also be a variable or record field.

The dynamic functions include:

- **lang_preference(lang)**
  This function returns the current language table selection. It takes as a parameter the current language value.
  You can use it to save new language settings in the user preferences for future reference. It forces the user interface to update its language table and repaint as necessary.
- **select(return_field, file_name, select_field, select_field_value, select_field_2, select_field_value_2)**
  This function works against many records to grab a scalar value from each one to make a list. You can use it to select just one record and use an array from within that record (for example, the `cap.exec` array field of the `operator` table) to make a display list.
  This function returns the value of a field (`return_field`) of a record that's from a table (`file_name`) and has a field (`select_field`) value equal to a specified value (`select_field_value`). The `select_field_value` can be a variable or a wildcard character (an asterisk). Both `select_field_2` and `select_field_value_2` are optional.
  The following are examples:
  - `select( "subcategory", "subcategory", "category", [$category] )`
  - `select( "name", "operator", "name", *)`
  - `select( "cap.exec", "operator", "name", [$operator] )`

Use dynamic functions

Use the following to select all the login IDs from a department to make a list:

```
select("name","operator","department",[$dept])
```

Use the following to select all the approvers for a certain group:

```
select("approvers","cm3groups","name",[$cm3group])
```

1.1.3. Format Control

Format Control allows the System Administrator to apply the following special processes to Service Manager files through individual forms:

- Validate fields.
- Establish user privileges.
- Display alternate record list forms.
- Do Calculations and Validations based on other fields in this file or other files.
- Call RAD routines.
- Define additional options and menu items.
- Automatically update or insert data in other parts of the database.

Format Control records can attach to any form or file within Service Manager and don't require special programming skills to implement. You define routines in Format Control that are user interactive or transparent and perform when a record opens or when a user adds, updates, or deletes a record from the database. Format Control is easy to apply and change.

While Format Control is a convenient utility, it should not be overused. Excessive reliance on Format Control to modify your system results in reduced system performance. If you intend to heavily customize your system, we recommend you find other, more efficient ways to implement your changes such as Data Policy or Data Validation.

Related topics

Tailoring
Format Control processes

1.1.3.1. Accessing Format Control

You can access the Format Control utility in Service Manager as follows:

Access Format Control from Forms Designer

Note: The best method for accessing Format Control is through Forms Designer. Accessing the record from the form with the same name as the Format Control record reduces your chances of accessing the wrong record.

1. Click **Tailoring** > **Forms Designer**.
2. Type the name of the form you want to view in the Form field of the Forms Designer dialog box.
3. Click **Search** to find the form.
4. Click **Format Control** from the More Actions menu. The Format Control record opens for the selected form.

Access Format Control from Database Manager

Note: To access the Format Control Additional Options for Change Management, you must select Administration Mode or you will be using Change Management instead of Database Manager.

Click **Tailoring** > **Database Manager**.

Access Format Control directly

Note: Accessing Format Control directly without going through Forms Designer enables you to search for an existing record or create a new record.

Click **Tailoring** > **Format Control**.

Open a Format Control record

1. Open Format Control using one of the following options:
   • Click **Tailoring** > **Format Control**.
   • Type `fc` in the Service Manager command line and click **Execute command** or press **Enter**. Service Manager opens a blank Format Control Maintenance form.
2. Type the name of the form or leave all fields blank, and then click **Search**.
   Service Manager opens a list of active forms. If only one form matches the query, Service Manager will display that form.
3. Click a form on the list to open it. Service Manager opens the selected form, showing the Main Information screen.

Add a Format Control record

1. Access Format Control from Forms Designer.
   Service Manager creates a new Format Control record for the form. Service Manager automatically places the form's name in the Name field. By default, the system automatically enters a value in the System field (a required field) and selects the Use Default Sort box.
   Note: When you access Format Control in this manner, the name of the new record defaults to the same name as the form used to access it. In the rare cases where you need a different name, override the default name.
2. Add the desired data.
3. Click **New** to add the record to the database.

Related topics

Format Control
Array maintenance
Array structure maintenance functions

1.1.3.2. Using expressions in Format Control

To create Format Control expressions you will need to be familiar with system functions, boolean (logical) fields and the file variable.

Use the following definitions when Creating Format Control expressions:

Primary File:
The file attached to a form. This is referred to as $file in a Format Control record. Additional files are defined as $file1, $file2, etc. in the order of their entry in the secondary files section of the Additional File Queries screen of the Format Control record.

Semicolon:
Semicolons(;) separate statements to allow them to be entered on one line.

Variable:
A value that resides in memory only, rather than in a database record. Variable names always start with $. To ensure that a variable contains the proper value, initialize it in the Initializations Section of the Format Control record.

Statements:
Service Manager supports some C-like statements.

- if then else
- do while
- for do

Format Control system functions

Many of the Service Manager system functions are used to create expressions in Format Control. The following is a list of the most commonly used functions:

Function	Description
denull()	Compresses an array by deleting all trailing NULL entries and returns the compressed array.
index()	Returns the index or position number for a specific element value in an array or character in a string. If the target value is not in the array or string, it returns 0 (zero).
lng()	Returns the number of elements in an array or structure and the number of characters in a string.
nullsub()	Replaces a null value with a known value, eliminating the possibility that expressions will return null or unknown results.
operator()	Returns the logon id of the current operator.
str()	Converts a non–string data type into a string.
substr()	Extracts a substring from a string.
tod()	Returns the current date and time.
val()	Converts a field into a numeric, logical, or date/time value.

Format Control Boolean (logical) fields

In the forms for Additional File Queries, Calculations, Validations, and Subroutines, you will find some or all of the following fields:

Conditional Field	Format Control processing area	Description
add	Additional File Queries, Calculations, Validations, JavaScript	Conditional expressions used in this field are evaluated before a record is added.
update	Additional File Queries, Calculations, Validations, JavaScript	Conditional expressions used in this field are evaluated before a record is updated.
delete	Additional File Queries, Calculations, Validations, JavaScript	Conditional expressions used in this field are evaluated before a record is deleted.
display	Additional File Queries, Calculations, Validations, JavaScript	Conditional expressions used in this field are evaluated when a screen/format is displayed or when the Format Control record is used with reports.
initial	Additional File Queries, Calculations, Validations, JavaScript	Conditional expressions used in this field are evaluated once, before a record is displayed the first time.
before	Subroutines	This field determines when the Conditional expressions in the add, update, and delete fields are evaluated. If the condition is true, the expression will be evaluated before the record is added, updated, or deleted. If the condition is false, the evaluated after the record is added, updated, or deleted. You can't use a variable in this field.

These fields must evaluate to true or false only. A value of true (or an expression that evaluates to true) in any of these fields causes the Format Control functions to execute. These functions execute before the edit is performed. If the fields are blank, the calculation or validation aren't performed for this table. For example, a value of true in the add field causes Format Control functions to be performed before a record is added. If the other fields (update, delete, display) are blank, the functions aren't performed when the record is updated, deleted, or displayed.

Normally, these fields contain true, false, blank, or they can contain a variable that's calculated in the Calculations process before being used in another process. When using either of these methods, use either the short or long version of Format Control to enter an expression that's evaluated to true or false at execution time.

This is an example of an expression that's frequently used:

```
index ("SysAdmin", $lo.ucapex)>0
```

This expression searches the user's capability array in the Operator Record to see if it has SysAdmin capabilities. The function executes if this evaluates to true, indicating that the operator has SysAdmin listed as a capability and therefore is authorized for the function.

Format Control file variable

The current database record (the record the user is opening for modifying) in Format Control is always identified by the variable $file. Elements of the current record being acted upon by Format Control associate with $file by expressions. For example, <field.name > in $file might identify a particular field in the current record to which a Format Control process is attached. The $file variable does NOT reference a file, which may use several forms to input data.

Important: In order for a property to depend on a variable or record field, the variable must be assigned as an input in Format Control.

Turn on menu forms by using Format Control

Applies to User roles: System Administrator

To turn on menu forms using Format Control:

1. Open Format Control using one of the following options:
   - Click **Tailoring** > **Format Control**.
   - Type `fc` in the Service Manager command line and click **Execute command** or press **Enter**.
     Service Manager opens a blank Format Control Maintenance form.
2. Type `login.DEFAULT` and then click **Search**.
3. Service Manager opens the login default form showing the Main Information screen.
4. Add `$G.show.menu.forms="true"` to the end of the **Initialization Expressions**.

Related topics

Format Control
Service Manager Programming

1.1.3.3. Sequential numbering for Format Control

Sequential numbering is a Subroutine process of Format Control that automatically adds identifying numbers to database records as they are created. Sequential numbering is used for Configuration Management, incident records, or employee records.

Sequential numbers are defined for appropriate fields in any data record. You create alphanumeric strings (prefixes and suffixes) that identify records by task or item.

Sequential numbering enables you to:

- Increment/decrement numbers in sequence.
- Reset values at definable intervals.
- Define values for increment/decrement sequence.
- Define the length of the number.
- Append prefixes and suffixes (characters) to sequential numbers.

The getnumb.fc application is called from the Subroutines process of Format Control to establish the sequential numbering sequence.

Related topics

Data types of sequential numbers
Create simple sequential numbers
Service Manager Programming

1.1.3.3.1. Data types of sequential numbers

There are two data types of sequential numbers:

- Simple numbers. For example, 1, 2, 3.
- Complex character strings. For example, (DEV00001/WS).

Note: If you want to use the prefix and suffix capabilities of sequential numbering, the field being incremented or decremented must be defined in the database dictionary as type character. In all other cases, the field must be of type number.

Related topics

Sequential numbering for Format Control
Common routines called from Format Control

1.1.3.3.2. Create simple sequential numbers

Applies to User roles: System Administrator

This example illustrates the process of setting up sequential numbering using Format Control.

To create simple sequential numbers:

1. Create the form for simple sequential numbers.
2. Create a database file for simple sequential numbers.
3. Create a Format Control record for simple sequential numbers.
4. Open the number file.
5. Add a numbered record to the database file for simple sequential numbers.

Create the form for simple sequential numbers[edit]

Applies to User roles: System Administrator

Create a form using the Form Wizard in Forms Designer called employees. The form will record simple employee data, such as name, address, and telephone number. Be sure to include a field labeled Employee ID with an input value of employee.id.

Create a database file for simple sequential numbers[edit]

Applies to User roles: System Administrator

To create a database file for simple sequential numbers:

1. Click **Tailoring** > **Forms Designer**.
2. Type employees in the form field, and click **Search**.
   Service Manager opens the employees form.
3. Click **Create File** in the More Actions menu.
4. Type employees in the filename field.
5. Click **OK** to create the new data file.
6. Change the Employee ID field of the employee file to a number data type.
   1. Select the employee.id field with the cursor.
   2. Click **Edit**.
   3. Click **number** in the Type field menu.

   Important: A data type of number is required for all simple sequential numbering.

Create a Format Control record for simple sequential numbers[edit]

Applies to User roles: System Administrator

To create a Format Control record for simple sequential numbers:

1. Click **Tailoring** > **Format Control**
2. Type employees in the Name field.
3. Click **New**.
4. Click **Subroutines** in the More Actions menu or click the **Subroutines** button.
5. Click **Show Expanded Form** in the More Actions menu.

6. Use the following values to complete the Subroutines process:

Application Name	Add	Before	Error Message
getnumb.fc	true	true	Sequential number did not process correctly!

Names	Definition	Values
record	The record in which the number will be placed. In this example, the employees data file.	$file
name	The name to appear in the Class field of the Number file generated by the Format Control record.	employees
prompt	The field within the record receiving the sequential number.	employee.id

7. Click **OK** to save the record.

Open the number file[edit]

Applies to User roles: System Administrator

To open the number file:

1. Click **Tailoring** > **Database Manager**.
2. Type number in the Form field of the Database Manager dialogue box.
3. Click **Search**.
4. When the blank Sequential Number File form opens, click **Search** to open a record list.
5. Select employees from the list of records. Service Manager opens the Sequential Number File for the employees file.

Add a numbered record to the database file for simple sequential numbers[edit]

Applies to User roles: System Administrator

To add a numbered record to the database file for simple sequential numbers:

1. Click **Tailoring** > **Database Manager**.
2. Type employees in the Form field.
3. Click **Search**.
4. Add data to the blank Employee Database record, but leave the Employee ID field blank.
5. Click **Add**.
   Service Manager adds the sequential number to the Employee ID field before the record is added to the database.

Contents

- 1 Create the form for simple sequential numbers
- 2 Create a database file for simple sequential numbers
- 3 Create a Format Control record for simple sequential numbers
- 4 Open the number file
- 5 Add a numbered record to the database file for simple sequential numbers

1.1.3.3.3. Create sequential number prefixes and suffixes

Applies to User roles: System Administrator

Sequential numbering is a subroutine process of Format Control that automatically adds identifying numbers to database records as they are created for Configuration Management, incident records, or employee records. Prefixes and suffixes are used to embed information in the sequential number and provide additional levels of identification. For example, each device in the database might be described by a logical name in the form of DEV(number)/(device type). For workstations, the device type might be WS. In this case, DEV is the prefix for the sequential number and /WS is the suffix.

For this example, see the task topics linked to below that will allow you to add a subroutine call to the device.workstation form to assign a unique device number to all records added to the file from this form.

To create sequential number prefixes and suffixes:

1. Create a Format Control record with prefixes and suffixes.
2. Add a record with prefixes and suffixes.

Create a Format Control record with prefixes and suffixes

Applies to User roles: System Administrator

Sequential numbering is a subroutine process of Format Control that automatically adds identifying numbers to database records as they are created for Configuration Management, incident records, or employee records. Prefixes and suffixes are used to embed information in the sequential number and provide additional levels of identification. For example, each device in the database might be described by a logical name in the form of DEV(number)/(device type). For workstations, the device type might be WS. In this case, DEV is the prefix for the sequential number and /WS is the suffix.

To create a format control record with prefixes and suffixes:

1. Click **Tailoring** > **Format Control**.
2. Type device.workstation in the Name field of the Format Control Initializations form.
3. Click **Search**.
4. Click **Subroutines** in the More Actions menu or click the **Subroutines** button.
5. Use the following values to complete the Subroutines process:

Application Name	Add	Before	Error Message
getnumb.fc	true	true	Sequential number did not process correctly!

Names	Definition	Values
record	The record in which the number is placed. In this example, the device.workstation record.	$file
name	The sequential number class to be used.	devices
prompt	The field within the record receiving the sequential number.	logical.name
string1	The string (prefix) to be added to the front of the sequential number when creating a new number.	DEV
query	The string (suffix) to be added to the end of the sequential number when creating a new record.	/WS
text	Defines the sequential number type (number or string). Since characters occur in the number, this must be string.	string
number1	Determines the length of the sequential number.	val("4", 1)

Add a record with prefixes and suffixes

Applies to User roles: System Administrator

Sequential numbering is a subroutine process of Format Control that automatically adds identifying numbers to database records as they are created for Configuration Management, incident records, or employee records. Prefixes and suffixes are used to embed information in the sequential number and provide additional levels of identification. For example, each device in the database might be described by a logical name in the form of DEV(number)/(device type). For workstations, the device type might be WS. In this case, DEV is the prefix for the sequential number and /WS is the suffix..

To add a record with prefixes and suffixes:

1. Click **Tailoring** > **Database Manager**.
2. Type device.workstation in the Form field.
3. Click **Search**.
4. Add data to the blank Workstation record.
5. Click **Add**.
   Service Manager adds a complex sequential number to the Asset field before the record is added to the database.

Notice that the complex sequential number DEV0002/WS appears in the Asset field (logical.name). Add another record, and the sequential number advances to DEV0003/WS.

Related topics

Sequential numbering for Format Control
Create simple sequential numbers

1.1.3.4. Array maintenance

Two types of arrays exist within Service Manager: simple arrays, consisting of fields of similar data types; and arrays of structures composed of related fields of similar or dissimilar data types. A series of applications exist in Service Manager that can be called from the Additional Options process of Format Control to perform various maintenance functions on

array structures. A single application called from the Subroutines process allows the user to sort simple arrays automatically when adding or updating records.

Related topics

Array structure maintenance functions
Add array structure maintenance options to a format control record
Test the array structure maintenance options I added
Service Manager Programming

1.1.3.4.1. Array structure maintenance functions

When using functions, such as copy, move, delete, and insert in the More Actions menu, these functions act against the entire array structure element, not against a single field within the structure.

The array structure editor reopens the field in question in an expanded, dynamic array field. The form that opens when starting the function is the same form that opens during execution. The routines that perform these functions are available collectively and individually to any RAD routine or Format Control Additional Options definition.

The following general rules apply to array structure maintenance:

- None of the routines perform update functions to the database. To save the changes, you must select the applicable update function within the application that called the maintenance functions.
- The applications are cursor dependent. For example, the insert function inserts a blank element at the cursor position within the array structure. If the cursor is placed in any field within an element of the array structure, the function is performed against that entire element.
- If the functions run against a non-array structure field, the procedure normally fails and either issues an invalid data-type message or produces unpredictable results.

The array structure maintenance RAD applications add buttons to the form that enable users to perform tasks, such as moving, copying, deleting or sorting lines in array structures.
They include:

- as.copy
- as.delete
- as.get.name
- as.insert
- as.move
- as.options
- as.sort

You can call the array structure maintenance RAD applications from the Additional Options Format Control process. You will need a thorough knowledge of Forms Designer and the use of subforms in the creation of array structures.

For more information on each application, a detailed explanation of how to define Additional Options, and how to pass data to these subroutines, see the related topics.

Note: RAD users should follow normal subroutine call procedures to run these applications.

Related topics

Format Control processes
Common routines called from Format Control
Array maintenance
Add array structure maintenance options to a format control record
Test the array structure maintenance options I added
Service Manager Programming

1.1.3.4.2. Add array structure maintenance options to a format control record

Applies to User roles: System Administrator

In this example we add the RAD application `as.options` to the format control record for the servicecontract form. The same general instructions work for any form.

To add array structure maintenance options to a format control record:

1. Click **Tailoring** > **Format Control**.
2. Type servicecontract in the Name field and press **Enter**.
3. If a Format Control record doesn't already exist for the form, click **New** to create one.
4. Click the **Addl Options** tab.
5. Type the following values into the appropriate fields. You can click **Show Expanded Form** from the More Actions menu if you prefer to work in it.

Field	Value
Option	2
Description	Array Options
Condition for Option	true
Reset on Return	true
Application	as.options
Error Message	Could not reset
Names	file name
Values	$file servicecontract

6. Click **Save**. Service Manager updates the Format Control record.

Related topics

Array maintenance
Array structure maintenance functions
Test the array structure maintenance options I added
Service Manager Programming

1.1.3.4.3. Test the array structure maintenance options that I added

Applies to User roles: System Administrator

In this example, we test the option added to Format Control record of the servicecontract form. The same general instructions work for testing any Format Control record.

To test the array structure maintenance options you added:

1. Click **Tailoring** > **Database Manager**.
2. Type servicecontract in the Form field and press **Enter**.
3. Select servicecontract from the list. Service Manager displays a blank Service Contract form.
4. Click **Search**.
5. Select a record from the list.
6. Click the **Named User** tab and insert your cursor in the array.
7. Click **Array Options** in the More Actions menu. The System Tray in the Service Contract form displays the buttons associated with the as.options subroutine.
    - Click **Insert** to insert a blank line in an array structure above the cursor position.
    - Click **Delete** to delete the line in which the cursor is placed.
    - Click **Copy** to copy the entry from the line selected by the cursor.

The original element is left in place when the copy is inserted into the array structure.
    - Click **Move** to copy the entry from the line selected by the cursor.
      The original element is deleted when the copy is inserted into the array structure.
      The System Tray display these buttons to complete the Copy and Move functions:
    - Insert inserts the copied element into the array structure above the line selected by the cursor.
    - Replace replaces the element selected by the cursor with the copied element.
8. Test the functionality of the buttons by inserting, copying, moving, and deleting named users.
9. Click **End** to restore the normal tray buttons.
10. Click **OK** or **Save** to save any changes you have made to the Service Agreement record.
    You must use **OK** or **Save** to update your record. Pressing **Enter** will NOT save any changes you have made.

Related topics

Array maintenance
Array structure maintenance functions
Add array structure maintenance options to a format control record

1.1.3.4.4. Sorting simple arrays

The sort.array application is a utility called from Format Control to sort simple arrays in either ascending or descending order. Supported array data types are:

- number
- character
- data/time

The sort.array utility is called from the Subroutines process of Format Control. Use this utility to sort simple arrays in ascending or descending order

To sort simple arrays, follow the procedures in these task topics:

1. Create the form for sorting simple arrays.
2. Create the Format Control record for sorting simple arrays.
3. Create a data record for sorting simple arrays.

Create the form for sorting simple arrays

Applies to User roles: System Administrator

To create a form for sorting simple arrays, use the Form Wizard in Forms Designer called demo.sort. It has three simple arrays of different data types: numbers, characters, and dates. Provide a Name field for the record and create a file in the database dictionary called demosort.

Use the following values to construct your form:

Field Name	Input Value
Name	name
Numbers	number
Characters	character
Dates	date

Create the Format Control record for sorting simple arrays

Applies to User roles: System Administrator

To create the Format Control record for sorting simple arrays, you will call the sort.array application from the Subroutines process of Format Control as follows:

1. Click **Tailoring** > **Format Control**

2. Type demo.sort in the Name field.
3. Click **New**.
4. Click **Subroutines** in the More Actions menu or click the **Subroutines** button.
5. Click **Show Expanded Form** in the More Actions menu.
6. Use the following values to complete the Subroutines process:

Application Name	Add	Update	Before	Error Message
sort.array	true	true	true	Could not sort the array.

Names	Definition	Values
file	When calling sort.array from Format Control, always pass $file to this parameter.	$file
name	The name of the input field (array) in $file you wish to sort.	number
boolean1	The parameter that controls the sort order. A value of true sorts in ascending order. If you pass false, the sort order is descending. The default is true.	val (“false”, 4) *

   For a detailed discussion of the val function, refer to System Language Help.
7. Click **OK** to save your record.

Create a data record for sorting simple arrays

Applies to User roles: System Administrator

Once you have created the form and Format Control for sorting simple arrays, you are ready to create the data record for sorting simple arrays.

To create a data record for sorting simple arrays:

1. Click **Tailoring** > **Database Manager**.
2. Type demo.sort in the Form field of the Database Manager dialogue box.
3. Click **Search**.
4. Type sort1 in the Name field of your test form.
5. Enter data in the arrays in random order. For example, numbers in the Numbers array and characters in the Characters array.
6. Click **Add** to add the record to the database and sort the array.
   Note: Since Update evaluates to true in the Format Control record for this form, you may also sort an array by clicking **Save**.
   The Numbers array is sorted in descending order.
7. Repeat the process for the other arrays by changing the name parameter in the Format Control record to reflect the array being sorted.
8. Remove the boolean1 parameter, and the sort order defaults to true. The array is sorted in ascending order.

Related topics

Format Control
Format Control processes

1.1.3.5. Determine parameters without using the RAD editor

Applies to User roles: System Administrator

You can determine the parameters for any application in Service Manager without using the RAD Editor. Use Database Manager to display the input value of each field in a particular application parameter panel. This value is the parameter name used in the Subroutines process of Format Control.

To determine parameters without using the RAD Editor:

1. Click **Tailoring** > **Database Manager**.
2. Type format in the Form field, and click **Search**.
3. Click **Format** in the record list.
4. Type the name of the application whose parameter values you want to view (for example, getnumb.fc) in the Format Name field of the blank Format form.
5. Click **Search**.
   The parameter Name opens in the Input field of the form and the function appears in the Label field.
6. Scroll down to view all the parameters listed.
7. Click **OK** and exit this record without saving it.

Related topic

Service Manager Programming Guide

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